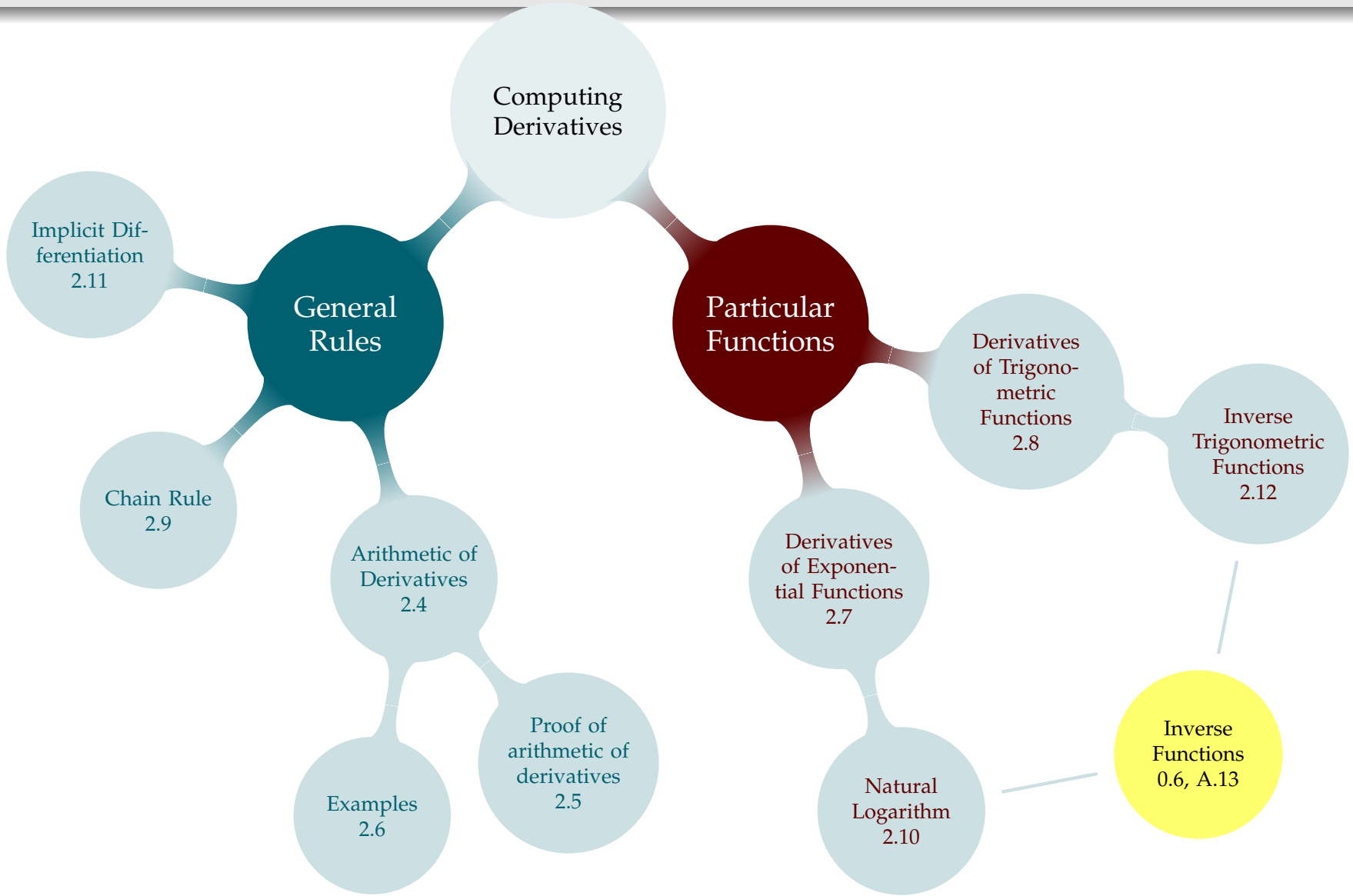
# Table of Contents

Skip review of inverse functions and logarithms



# Invertibility Game

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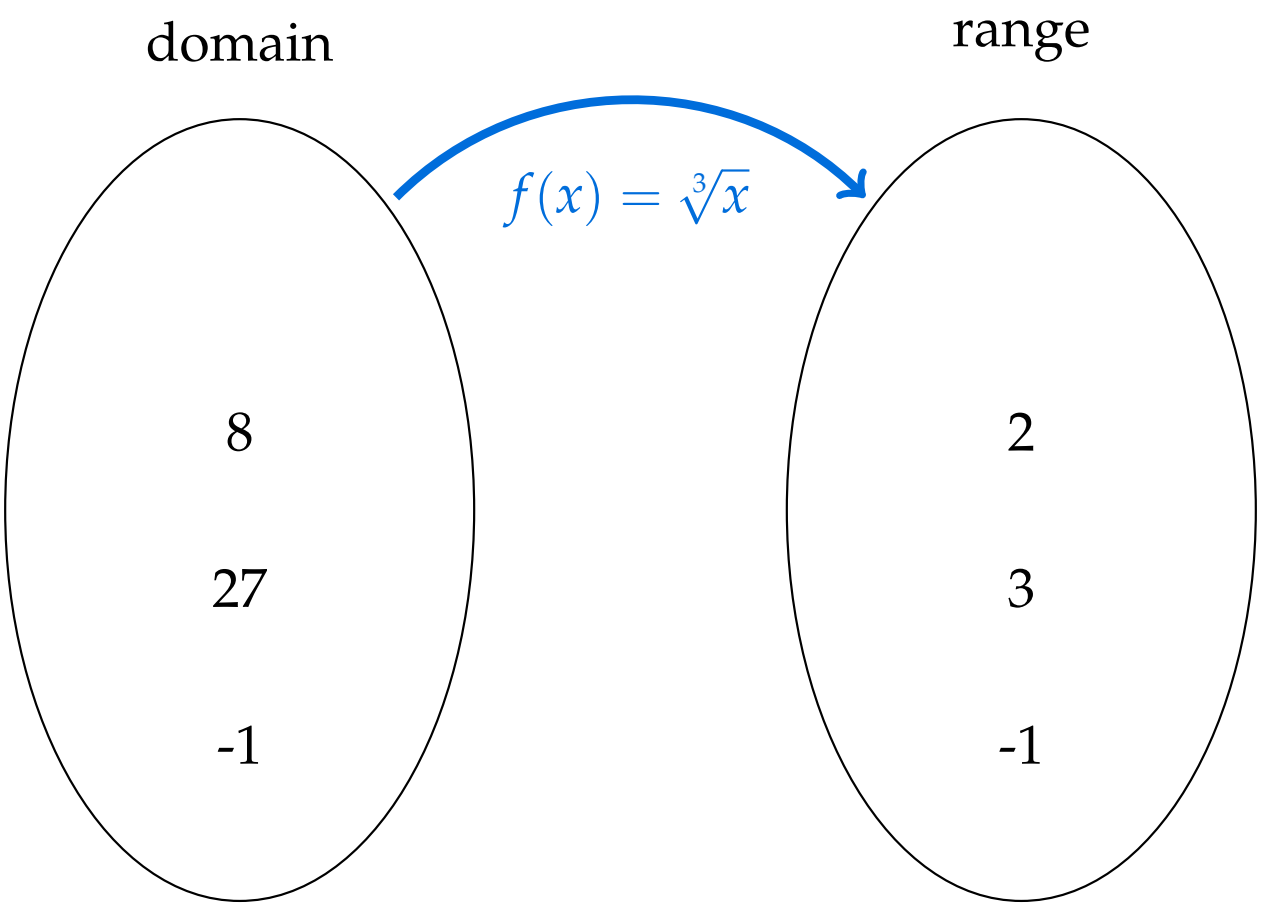
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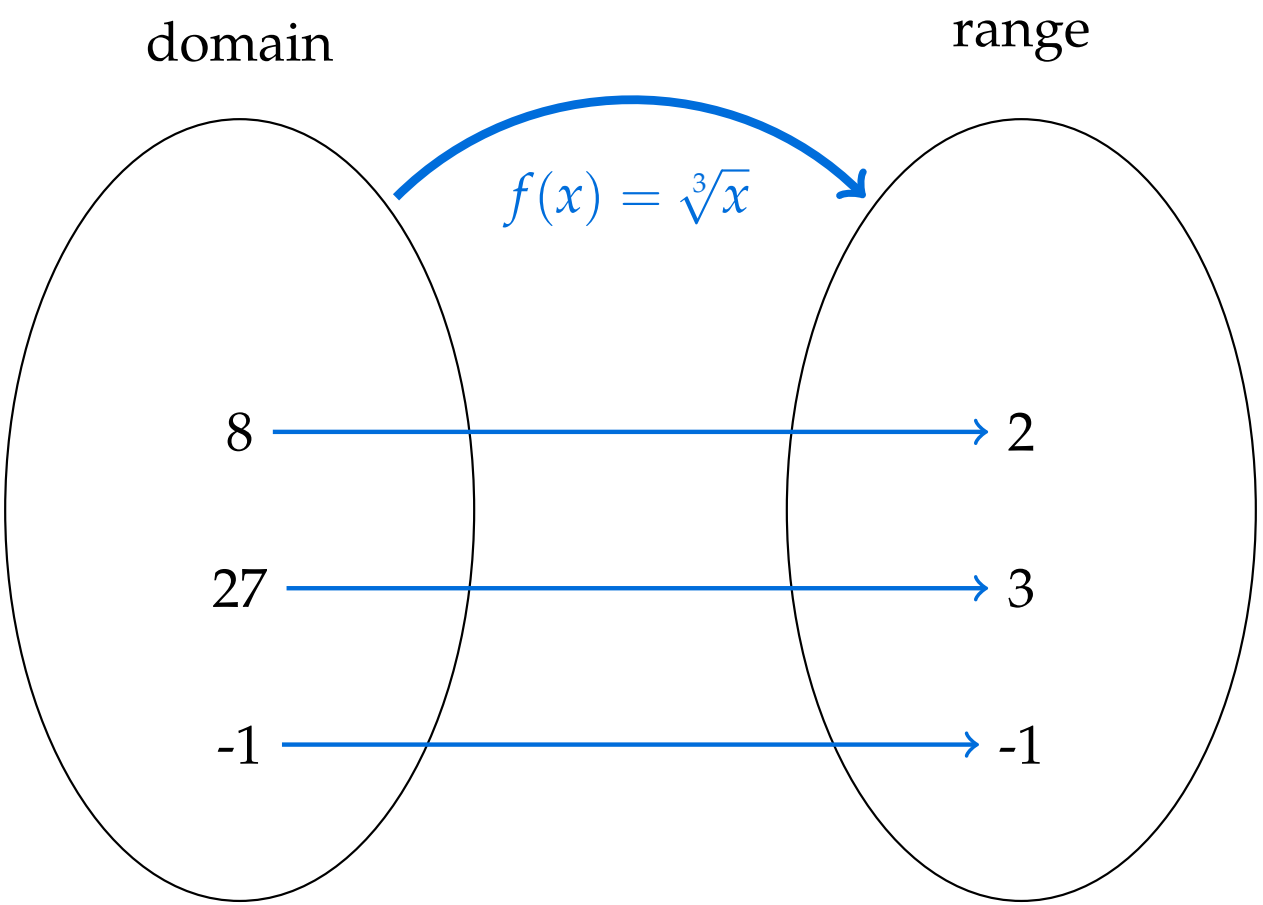
**Round 3:** $f(x) = |x|$

**Round 4:** $f(x) = \sin x$

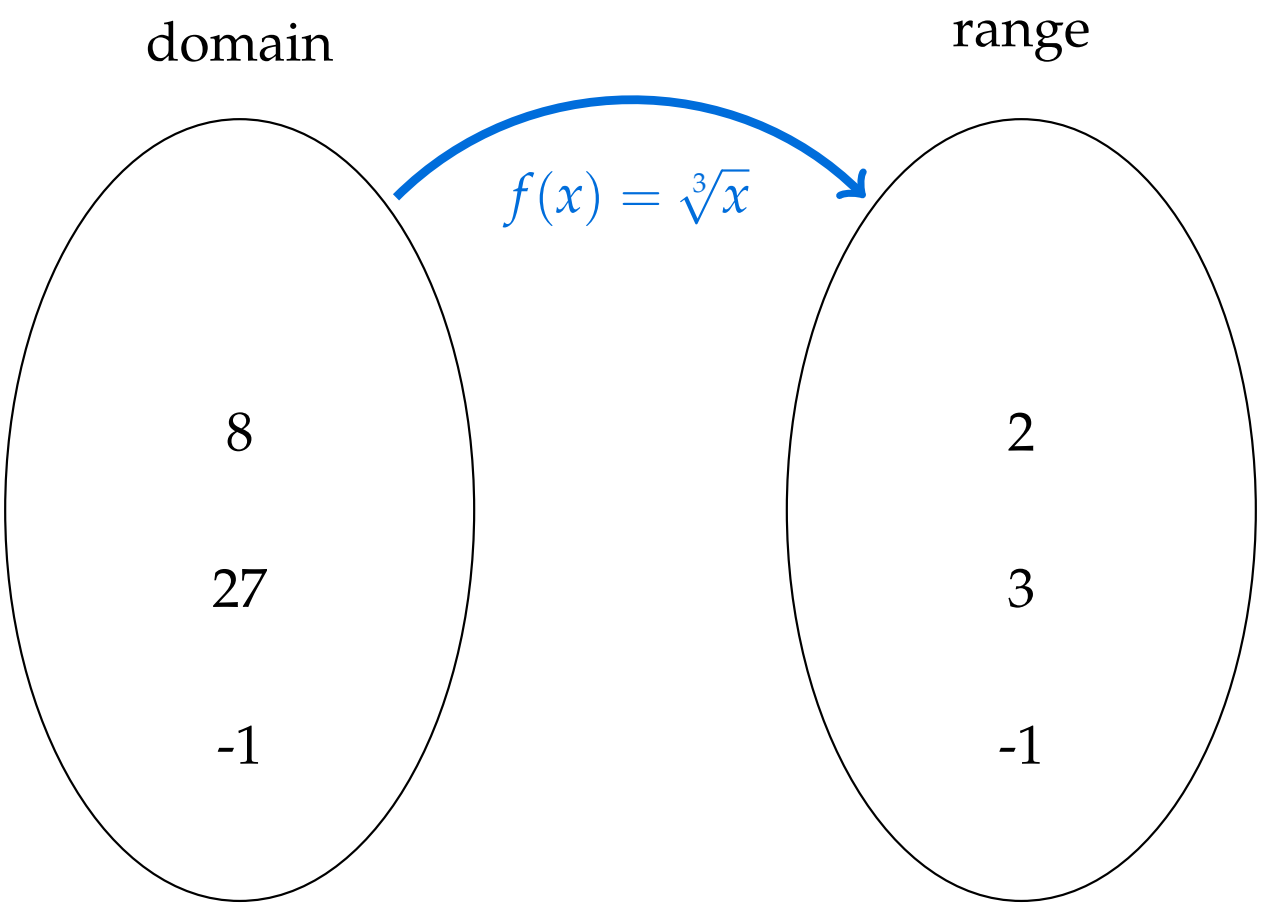
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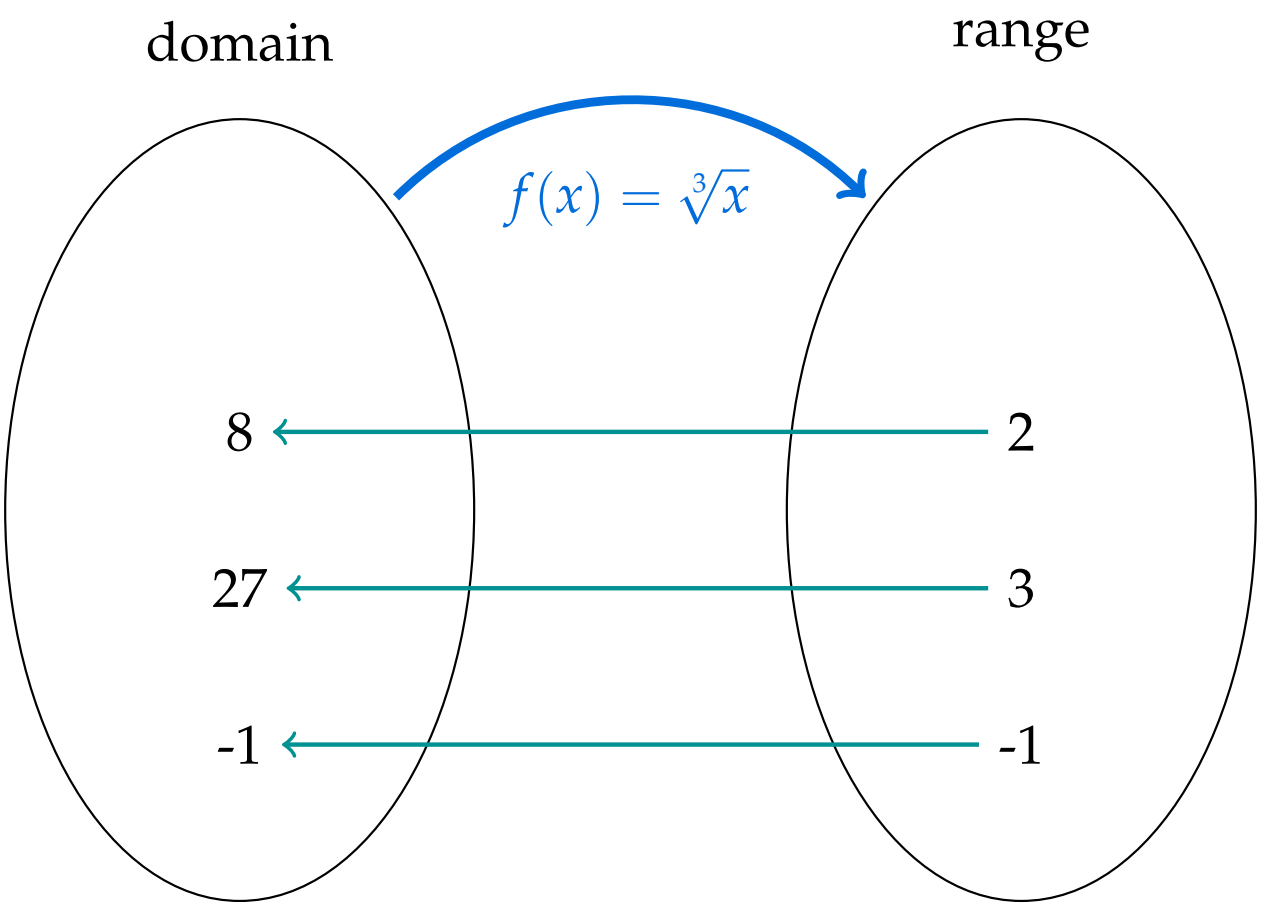
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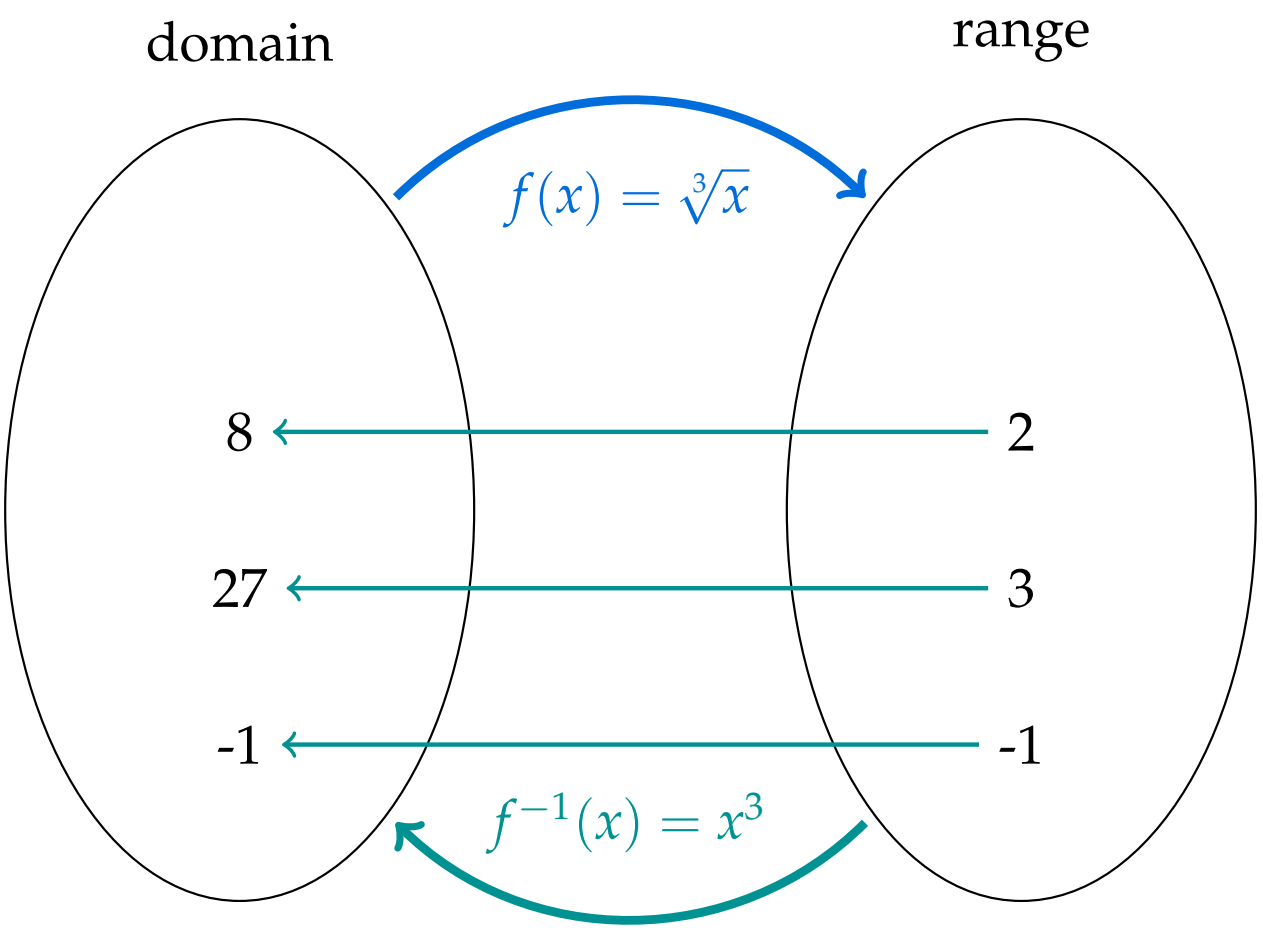
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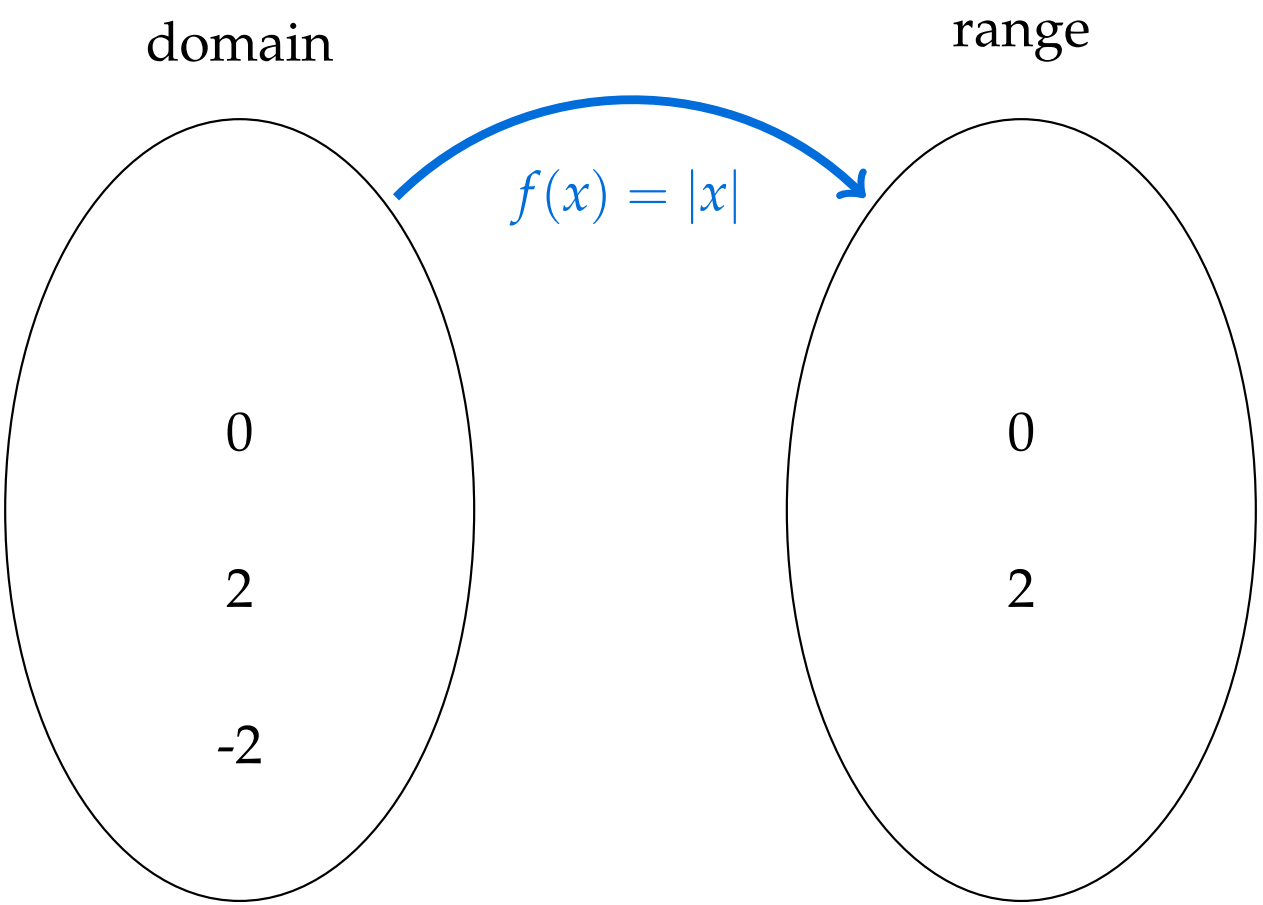
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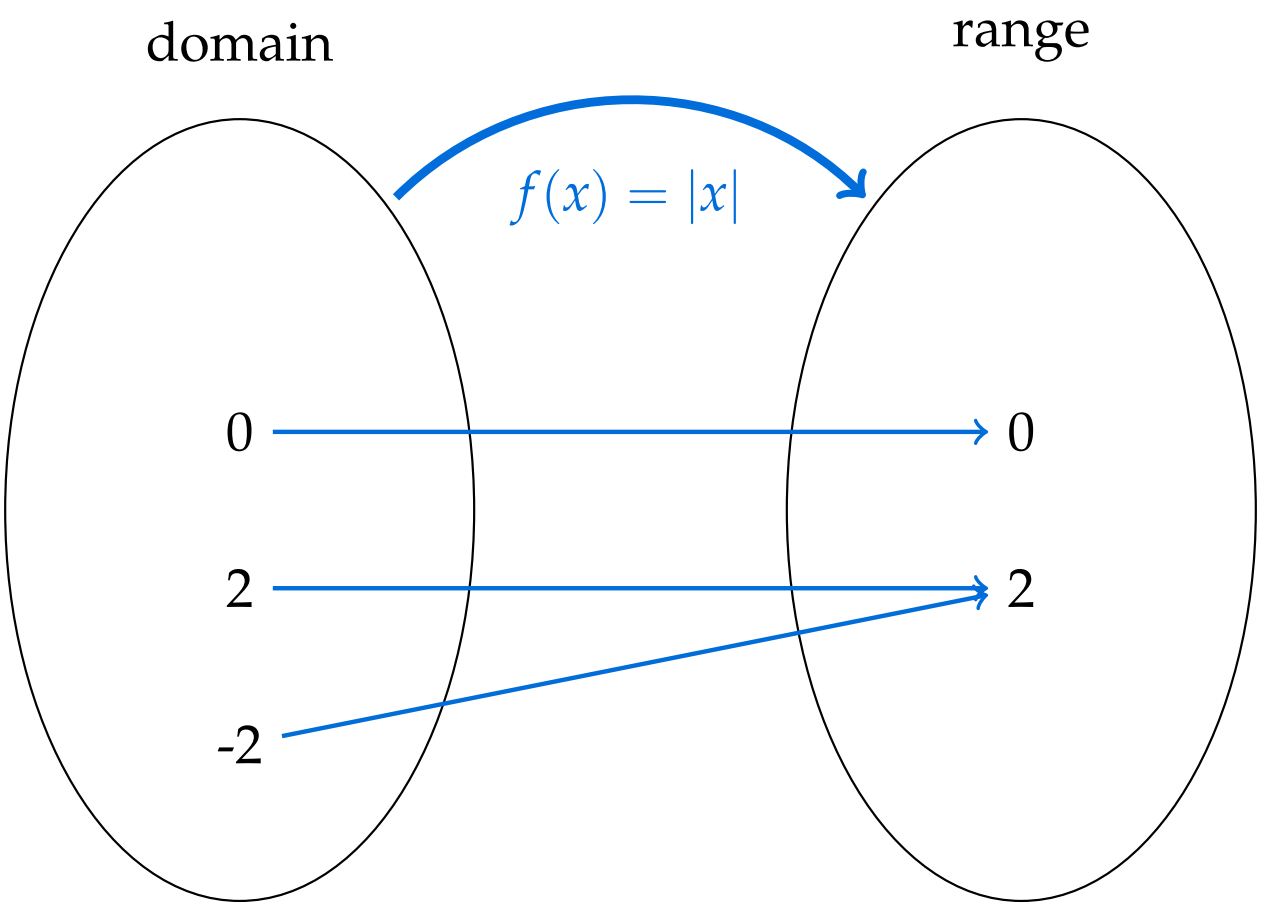
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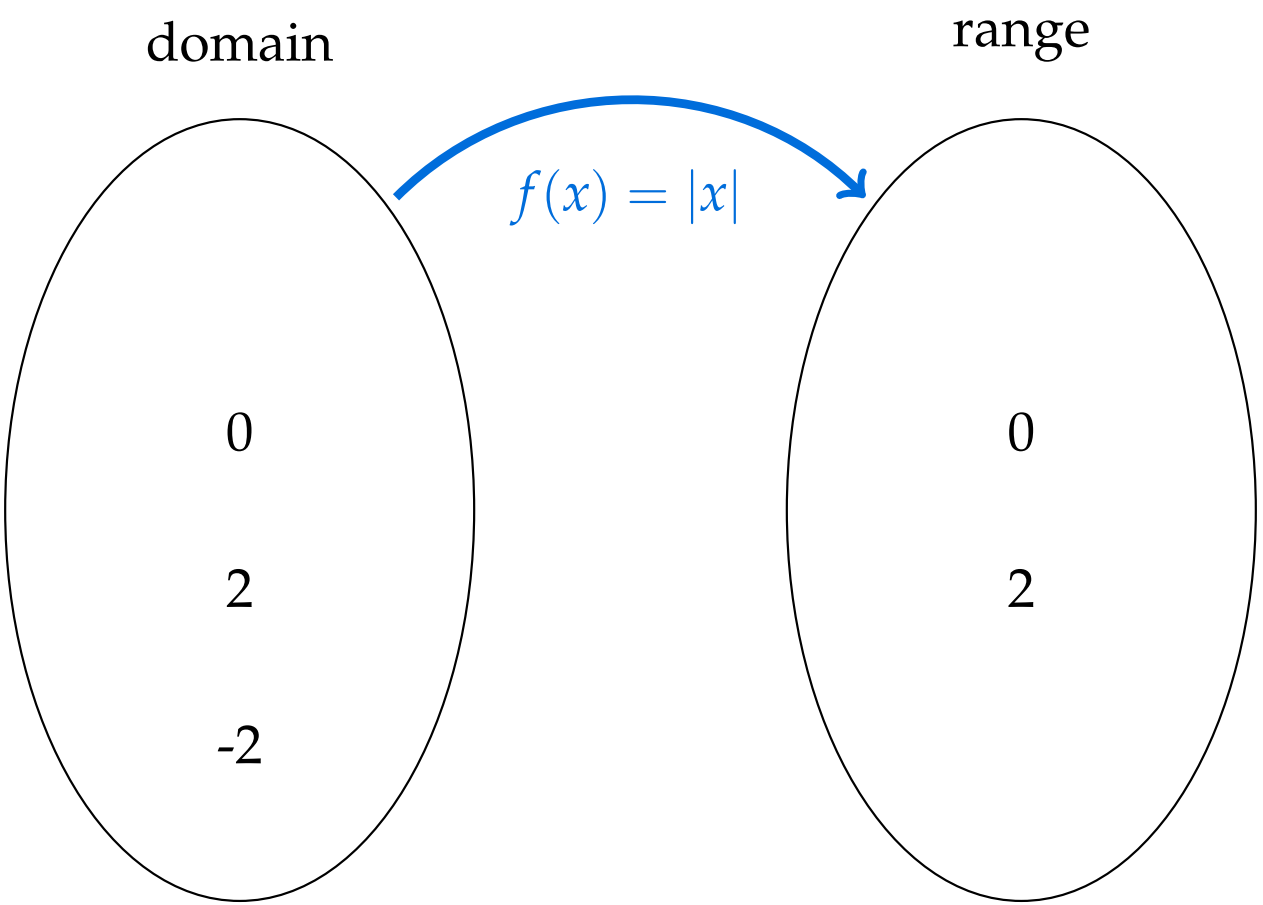
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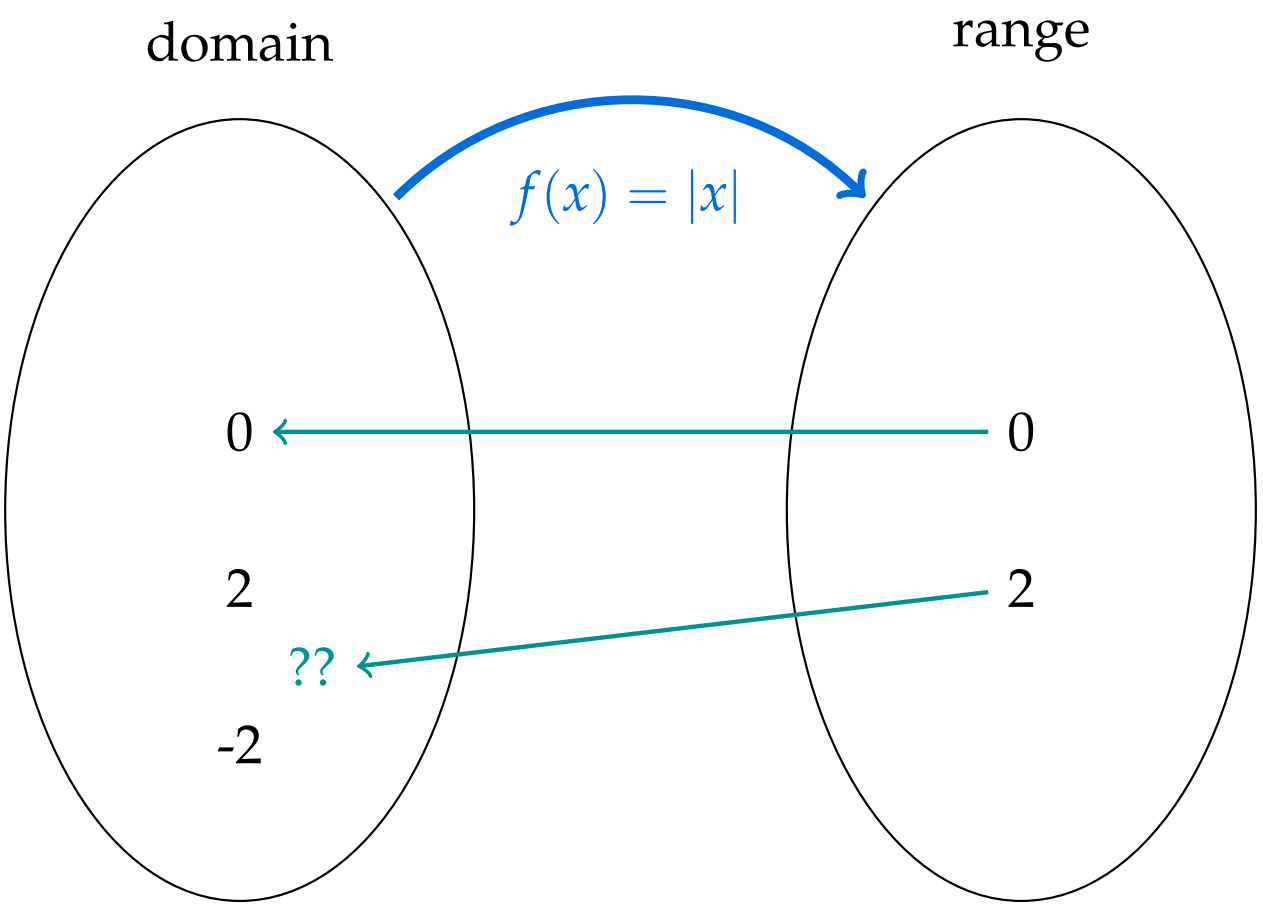
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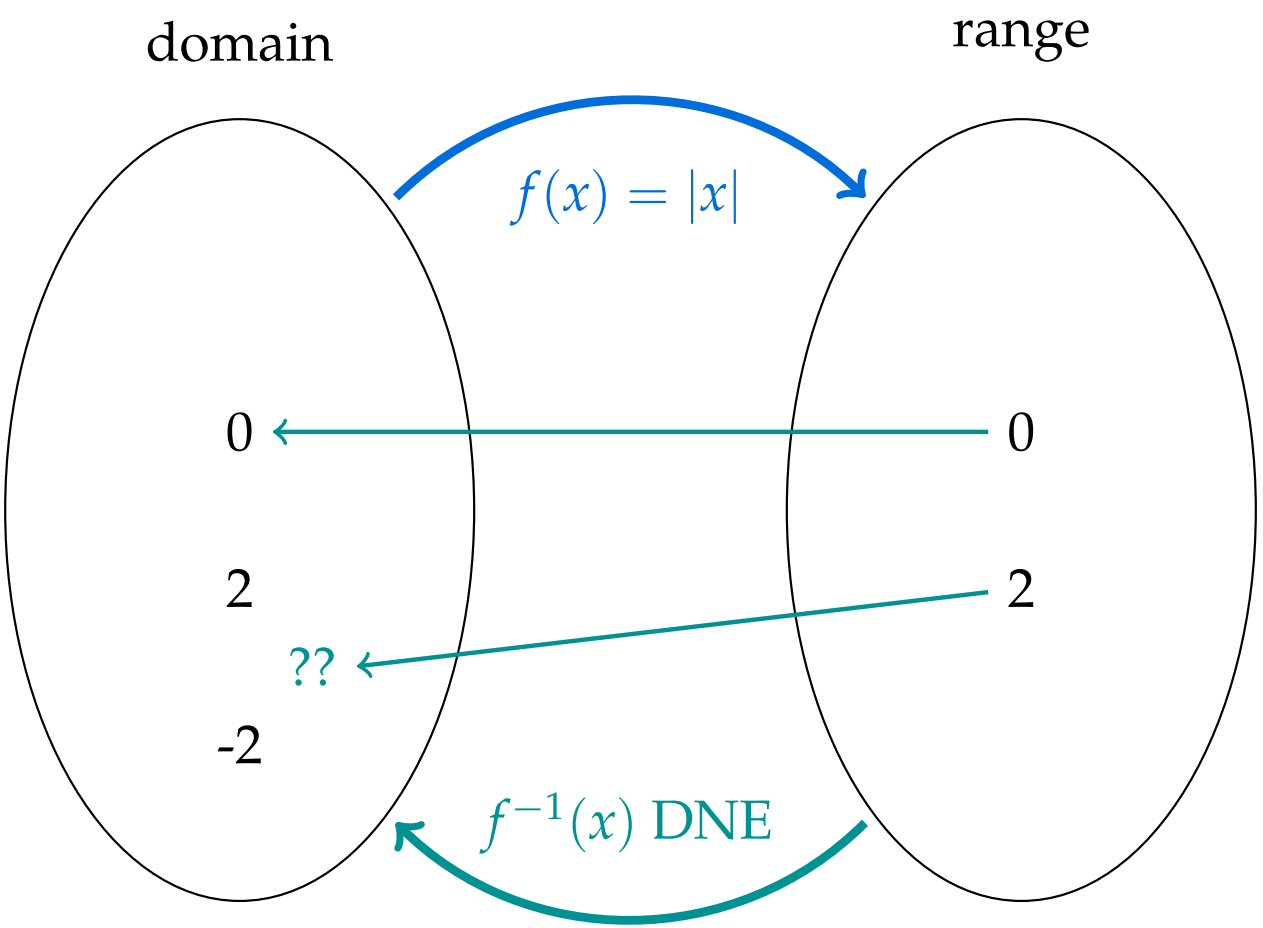
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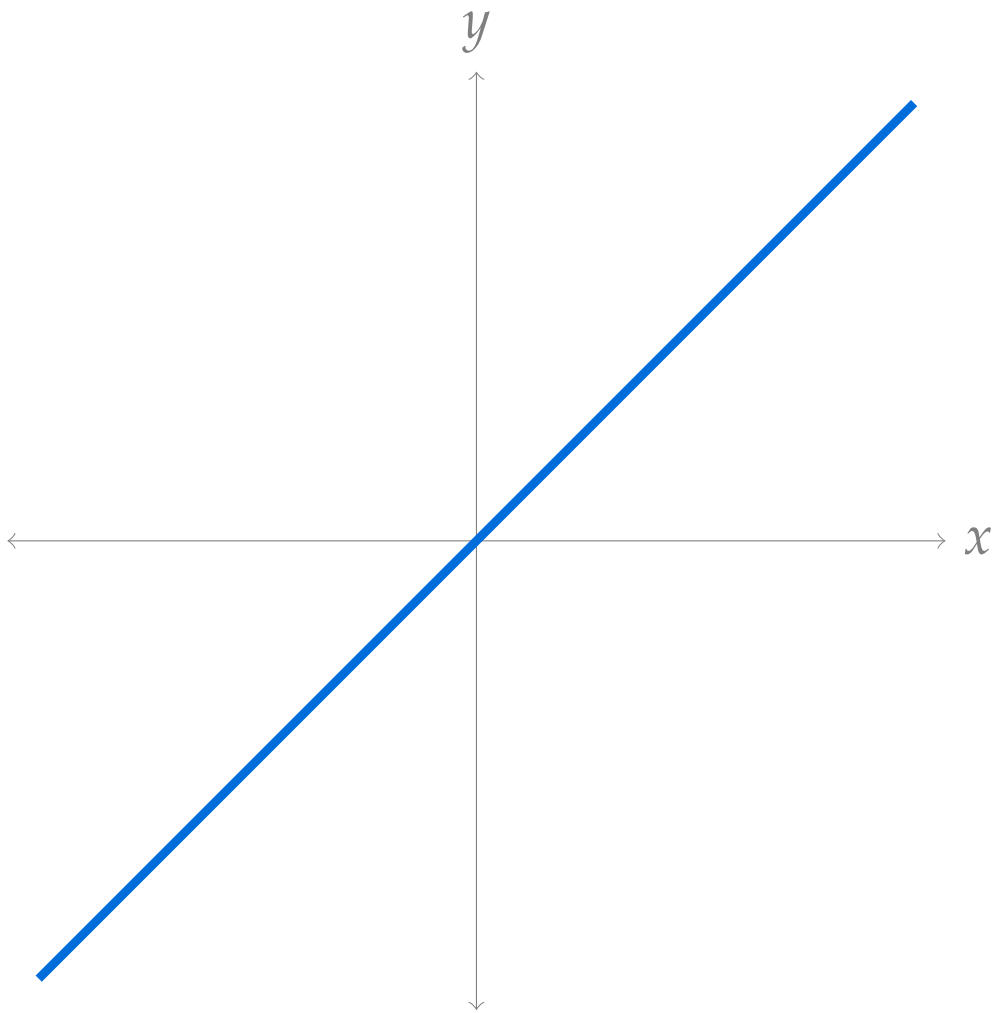


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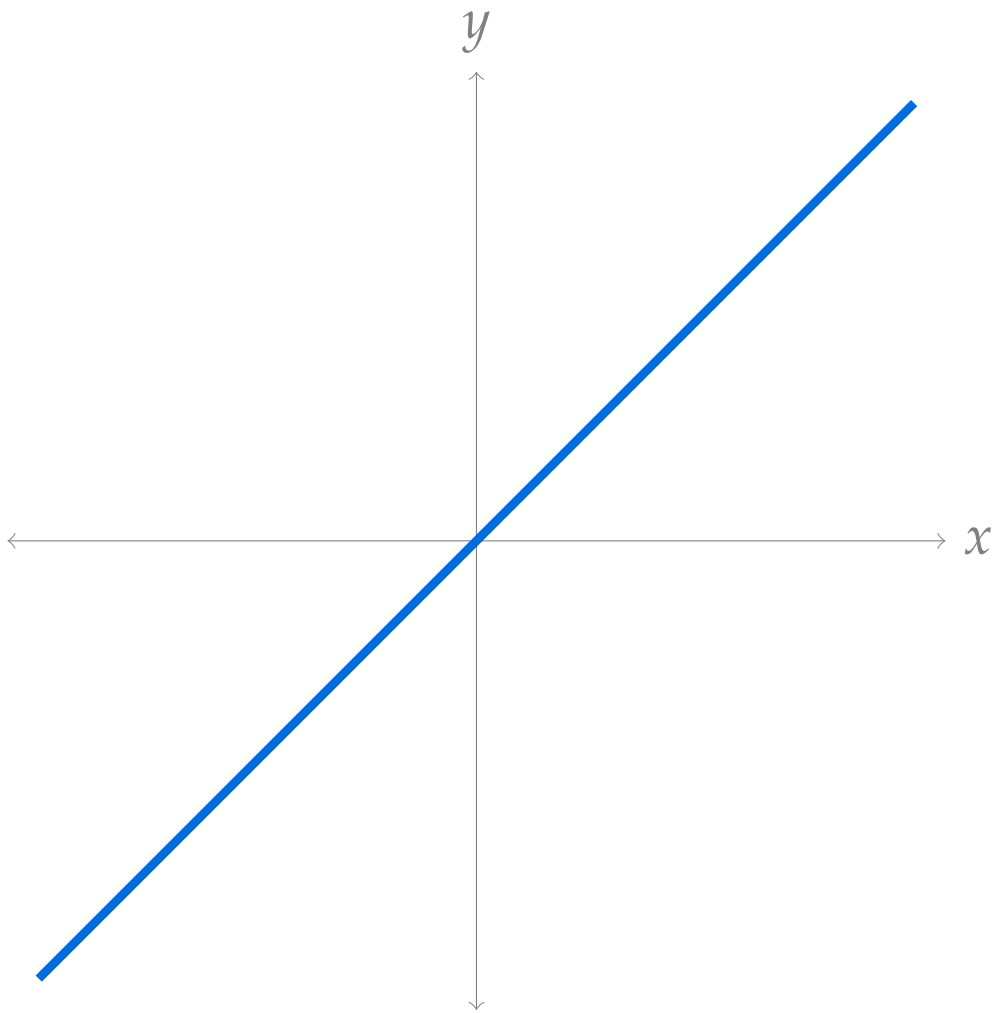
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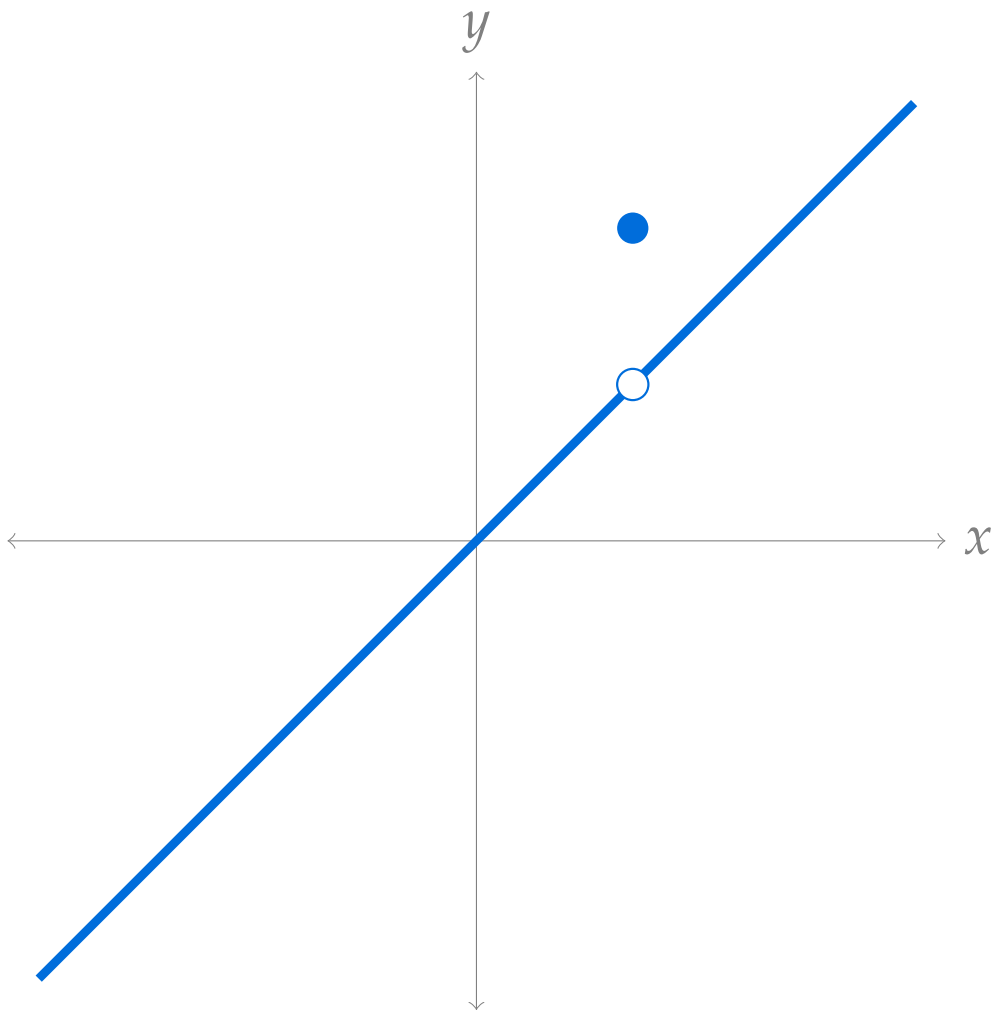

A. invertible

B. not invertible



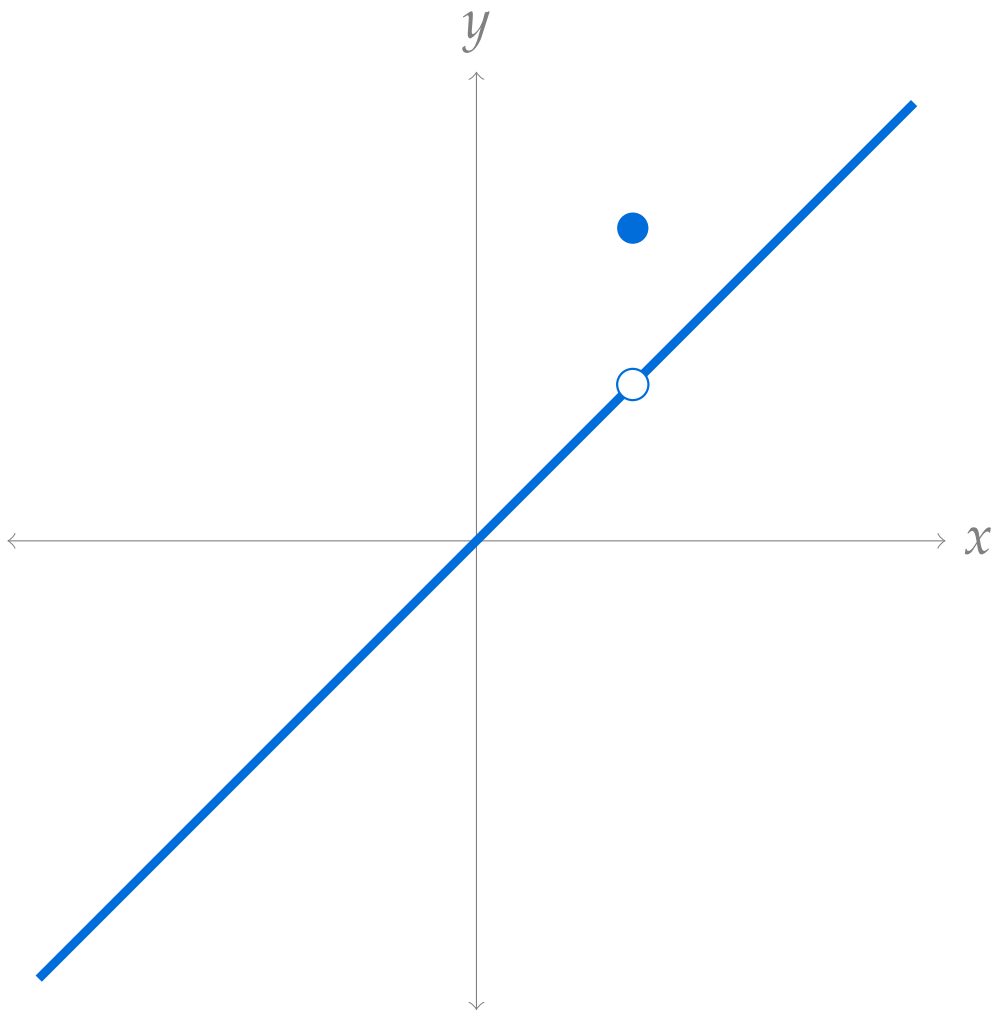

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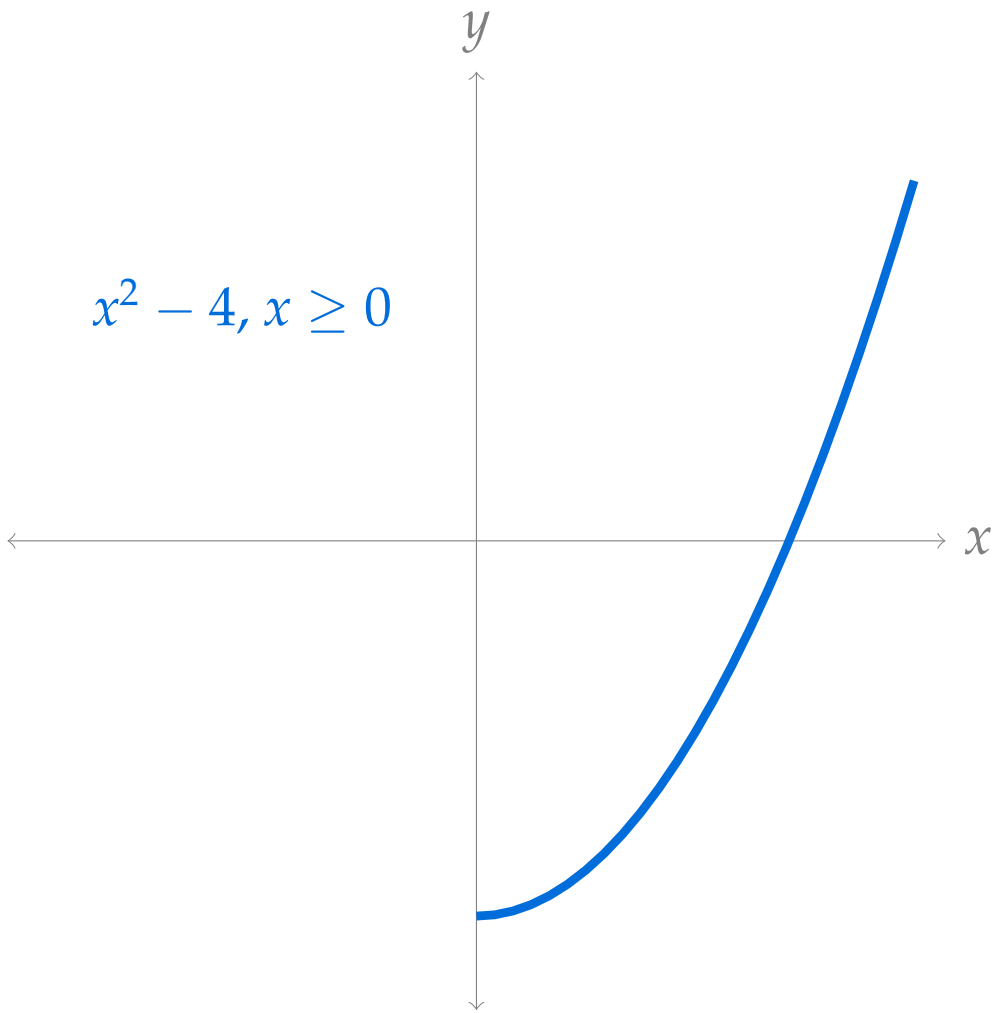

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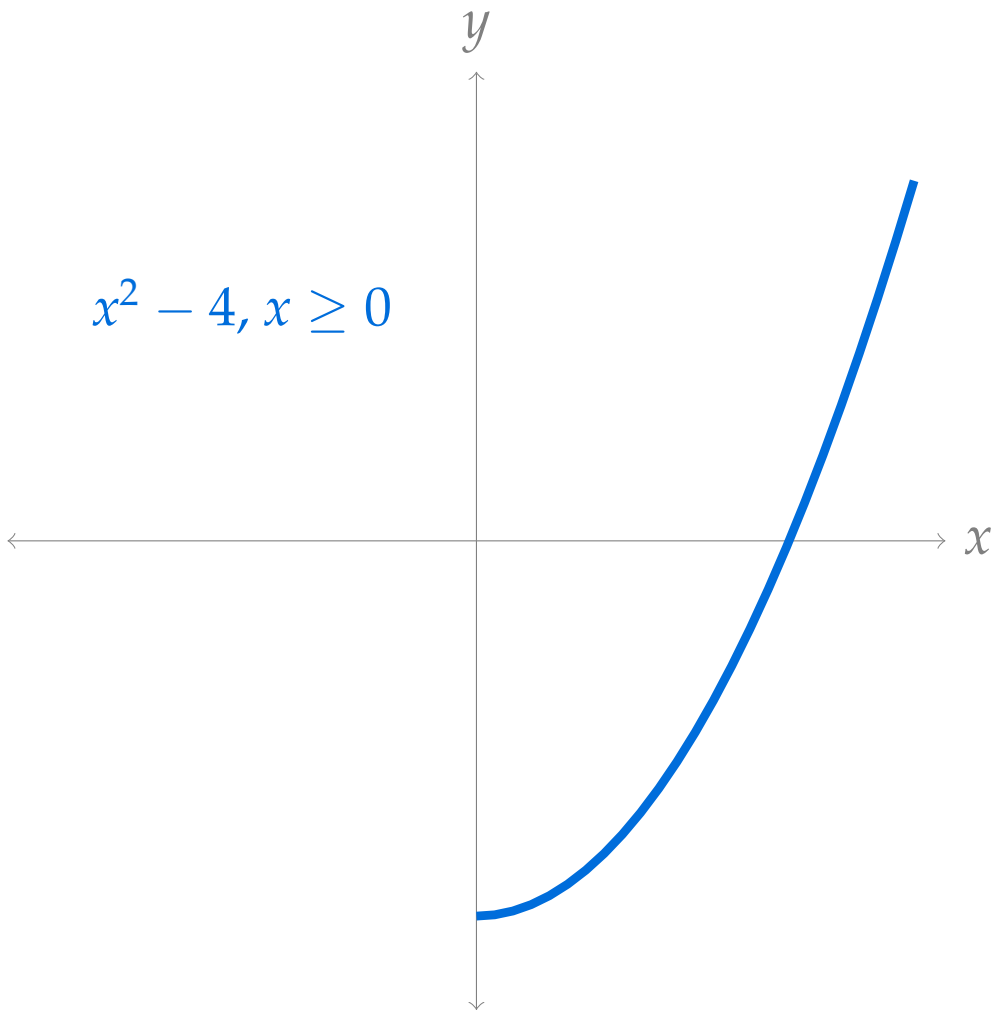

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# Relationship between $f(x)$ and $f^{-1}(x)$

Let $f$ be an invertible function.
What is $f^{-1}(f(x))$?

A. $x$

B. 1

C. 0

D. not sure

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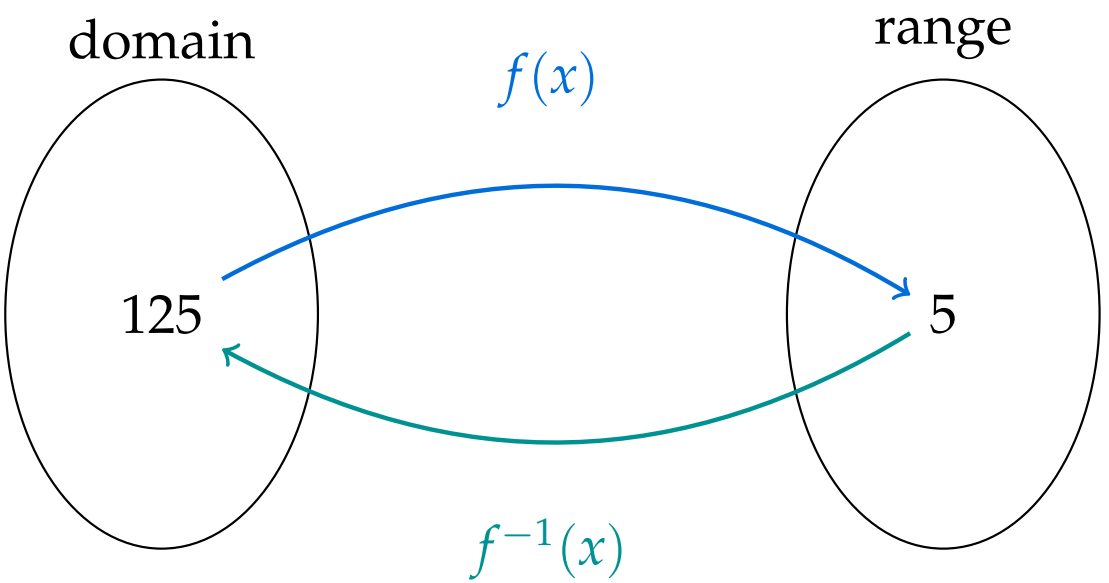
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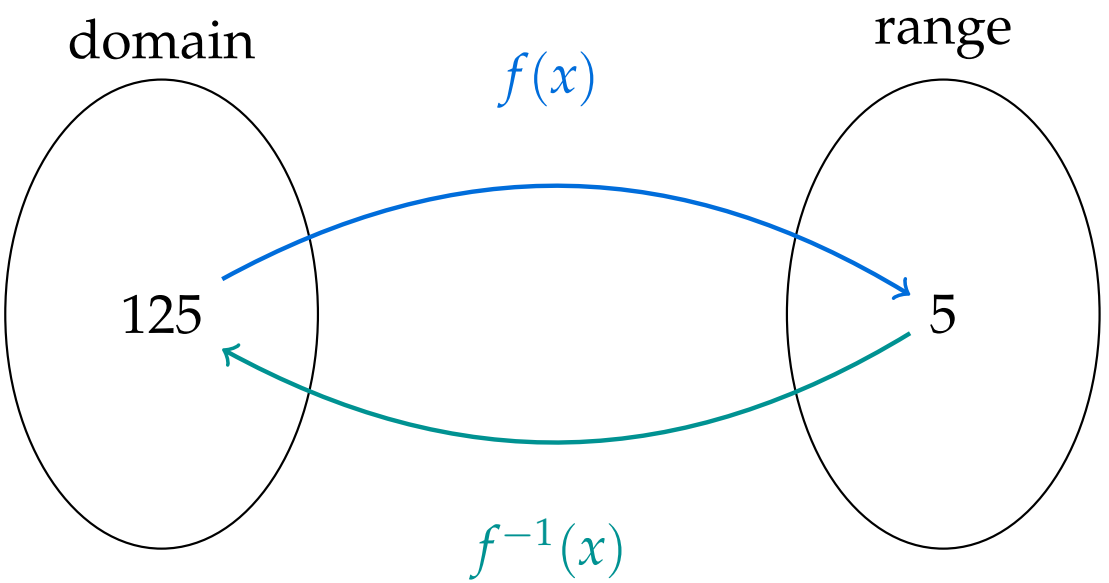
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What is $f^{-1}(10)$? (do not simplify)

Q

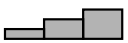
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What is $f^{-1}(x)$?



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In order for a function to be invertible , different $x$ values cannot map to the same $y$ value.
We call such a function **one-to-one**, or **injective**.

Suppose $f(x) = \sqrt[3]{19 + x^3}$. What is $f^{-1}(3)$? (simplify your answer)
$f(2) = 3$, so $f^{-1}(3) = 2$

What is $f^{-1}(10)$? (do not simplify)
$\sqrt[3]{19 + y^3} = 10$ tells us $f^{-1}(10) = \sqrt[3]{10^3 - 19}$

What is $f^{-1}(x)$?
$\sqrt[3]{19 + y^3} = x$ tells us $f^{-1}(x) = \sqrt[3]{x^3 - 19}$

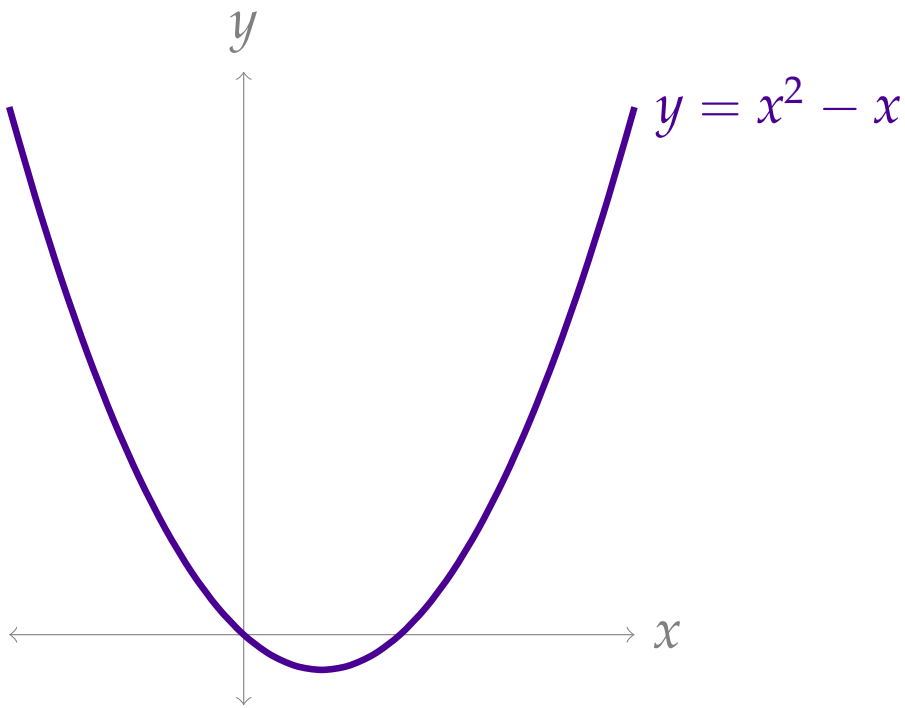
Let $f(x) = x^2 - x$.

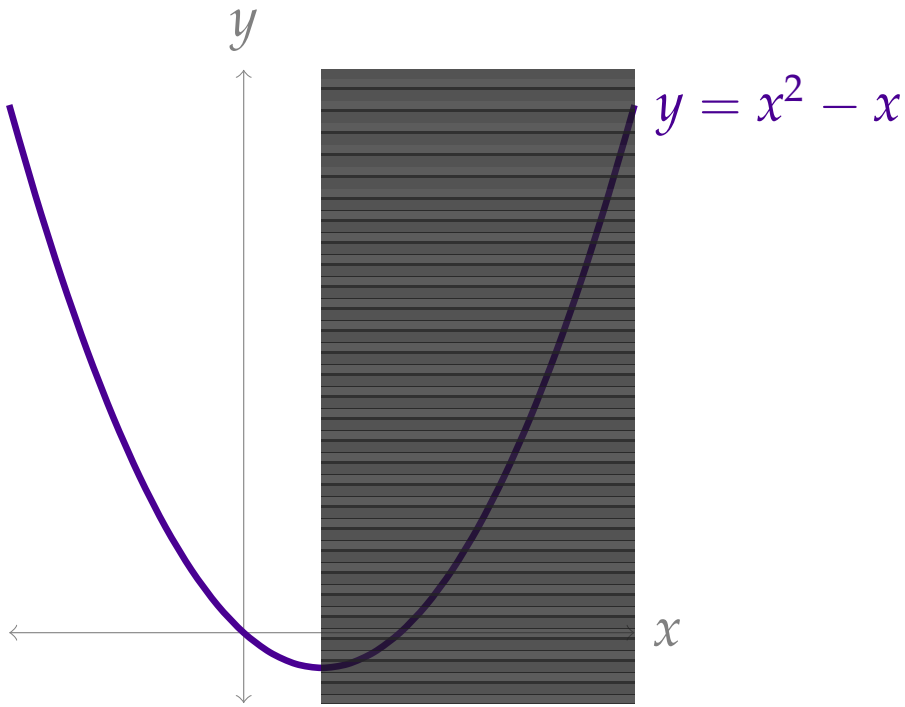
1. Sketch a graph of $f(x)$, and choose a (large) domain over which it is invertible.

2. For the domain you chose, evaluate $f^{-1}(20)$.

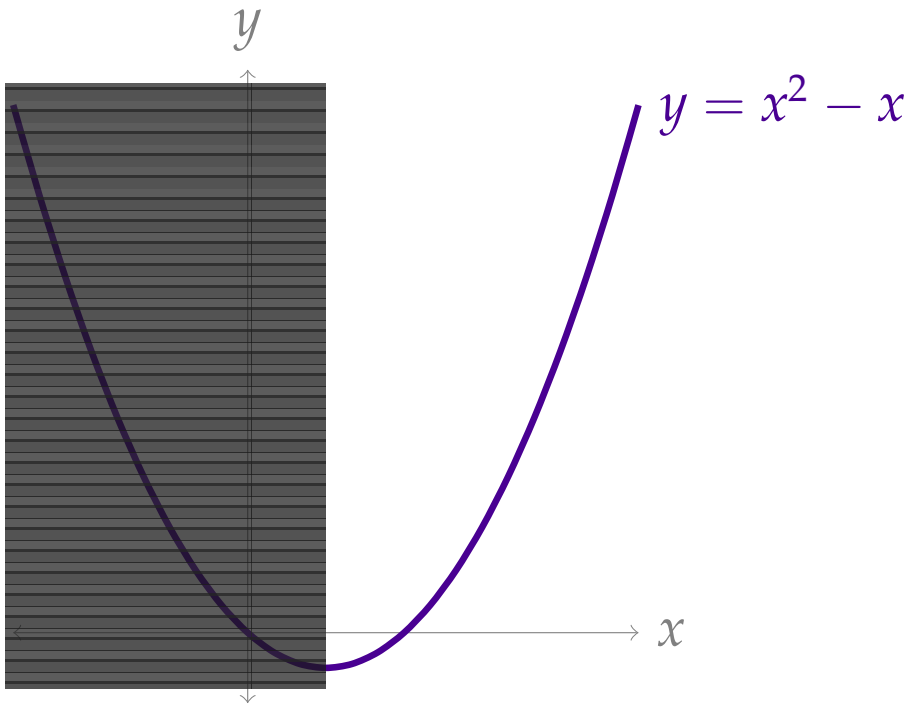
3. For the domain you chose, evaluate $f^{-1}(x)$.

4. What are the domain and range of $f^{-1}(x)$? What are the (restricted) domain and range of $f(x)$?

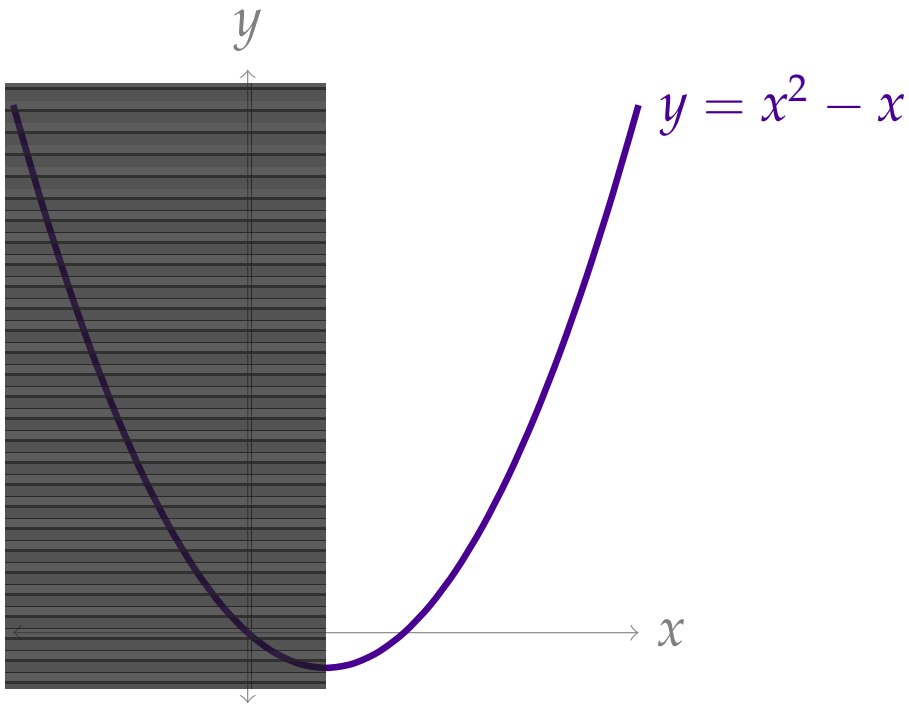

y
$y = x^2 - x$
x




Domain: $\left(-\infty, \frac{1}{2}\right]$

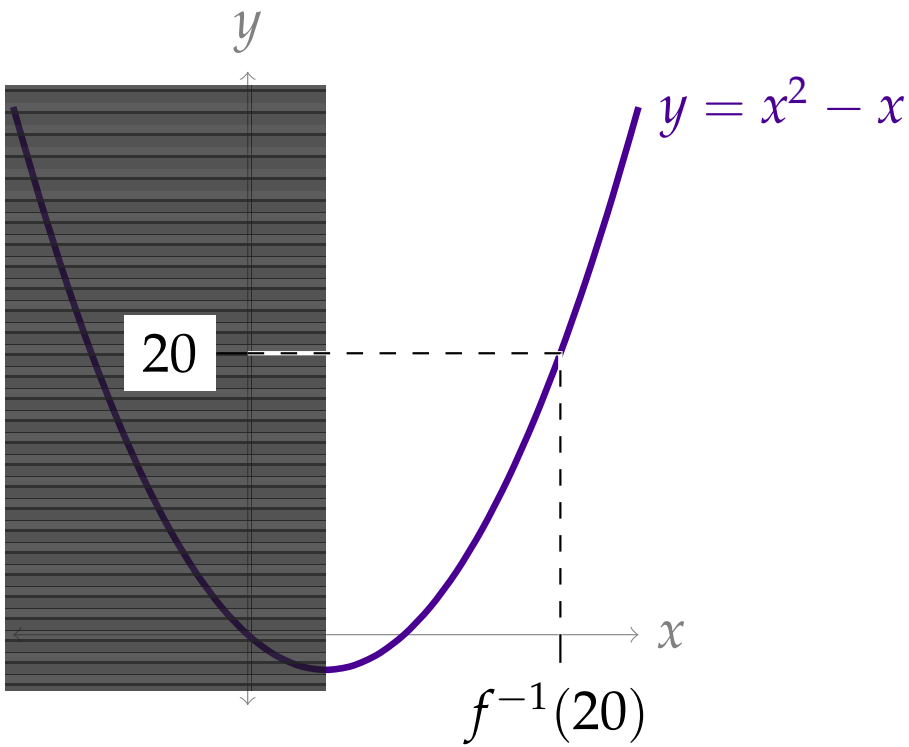



Domain: $\left[\frac{1}{2}, \infty\right)$

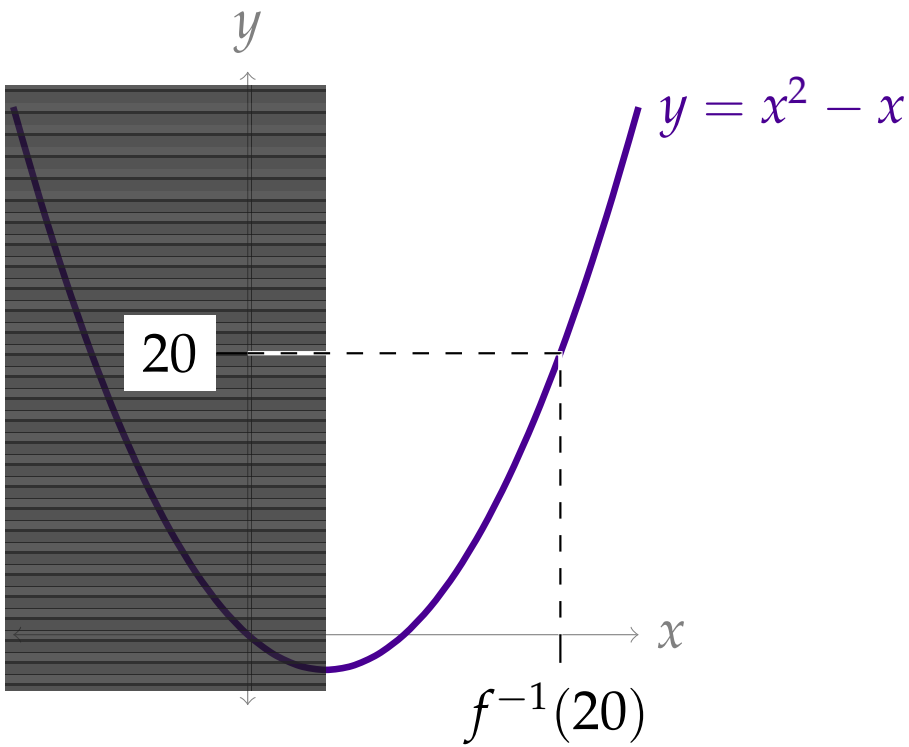



Domain: $\left[\frac{1}{2}, \infty\right)$

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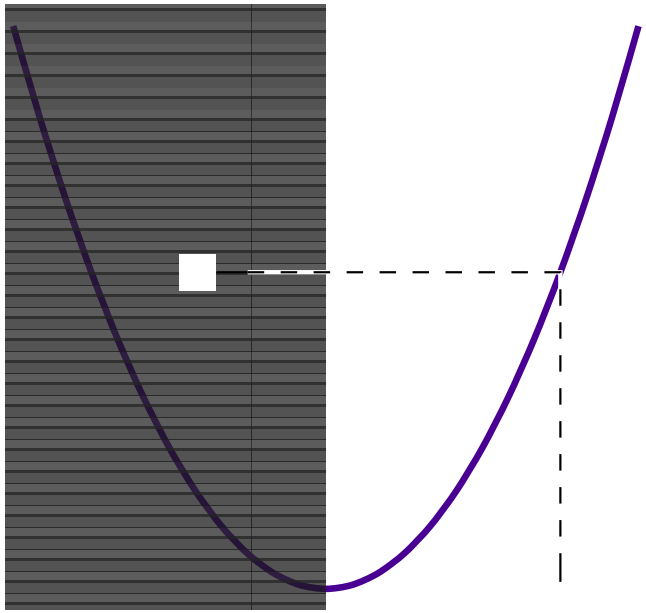



$f^{-1}(20) = 5$

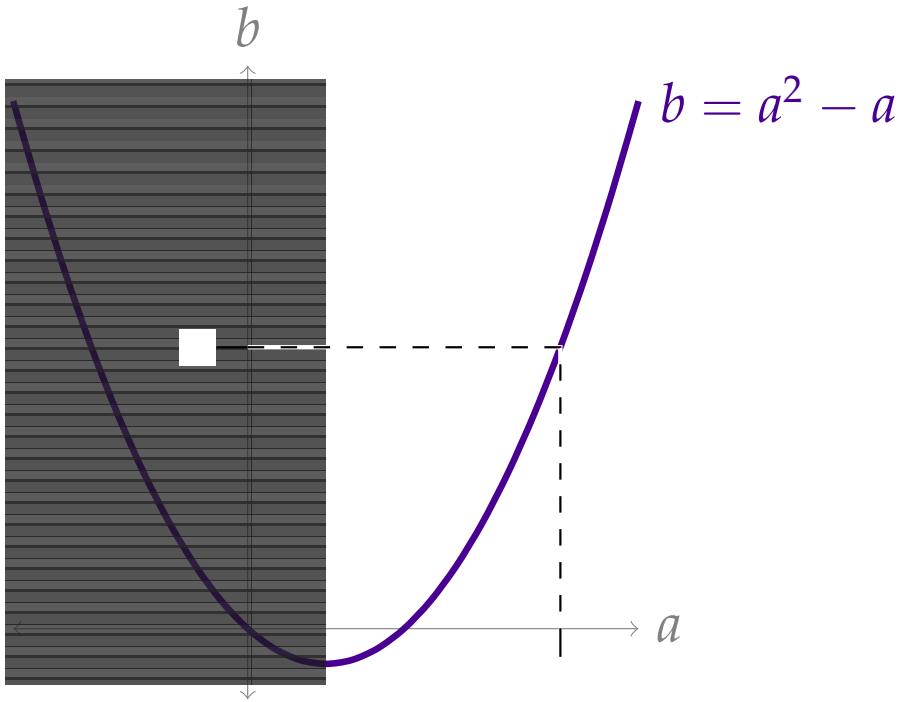
$$20 = x^2 - x$$

$$0 = x^2 - x - 20 = (x-5)(x+4)$$

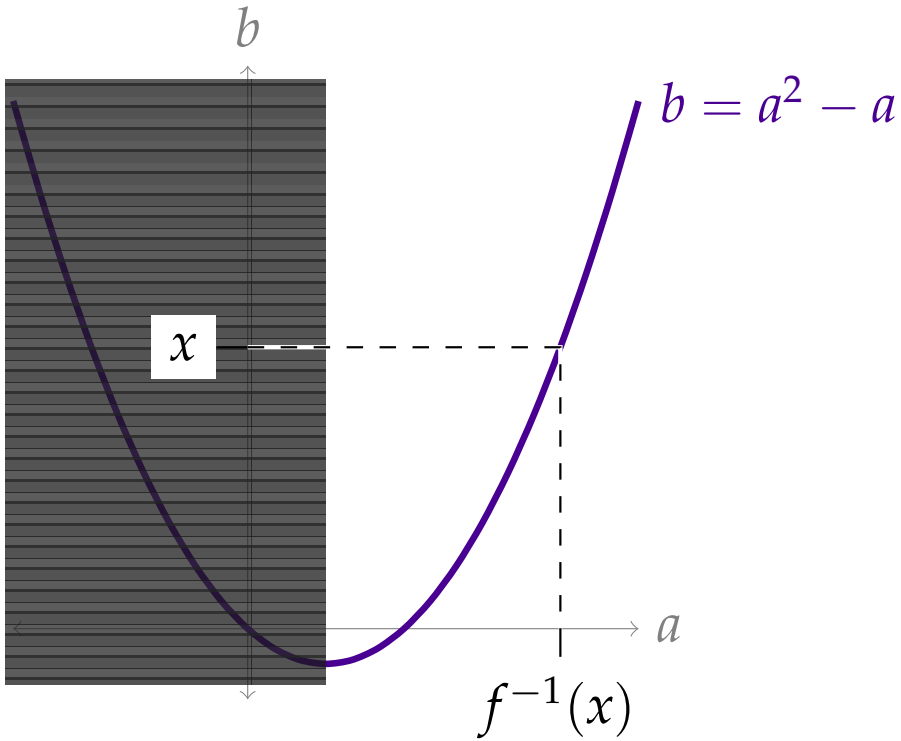
$$x = 5$$



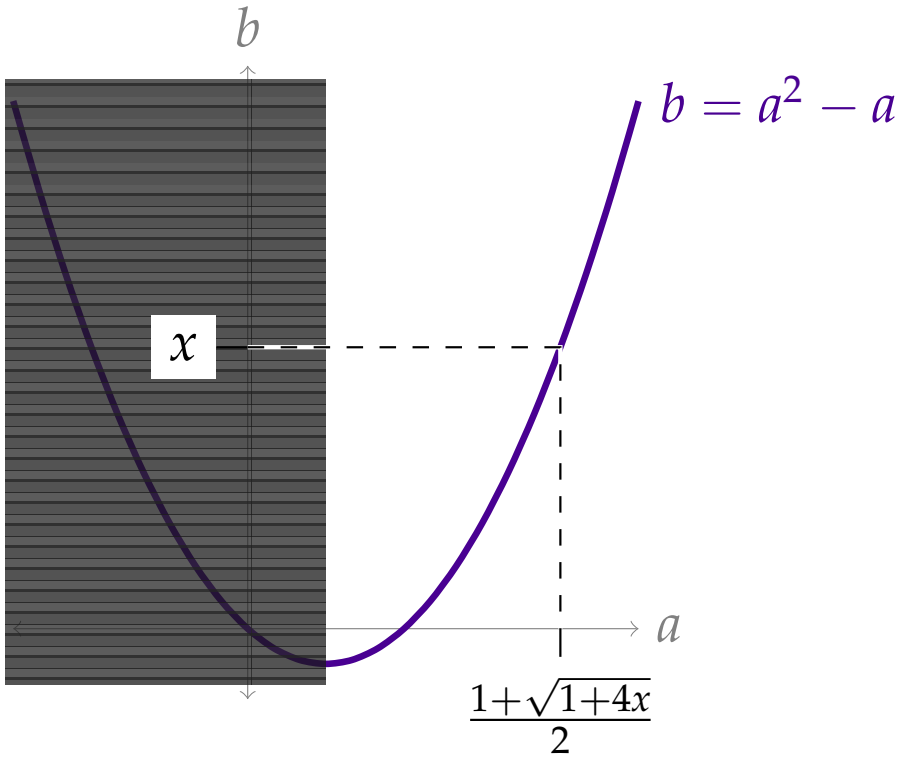
$$f^{-1}(x) =$$




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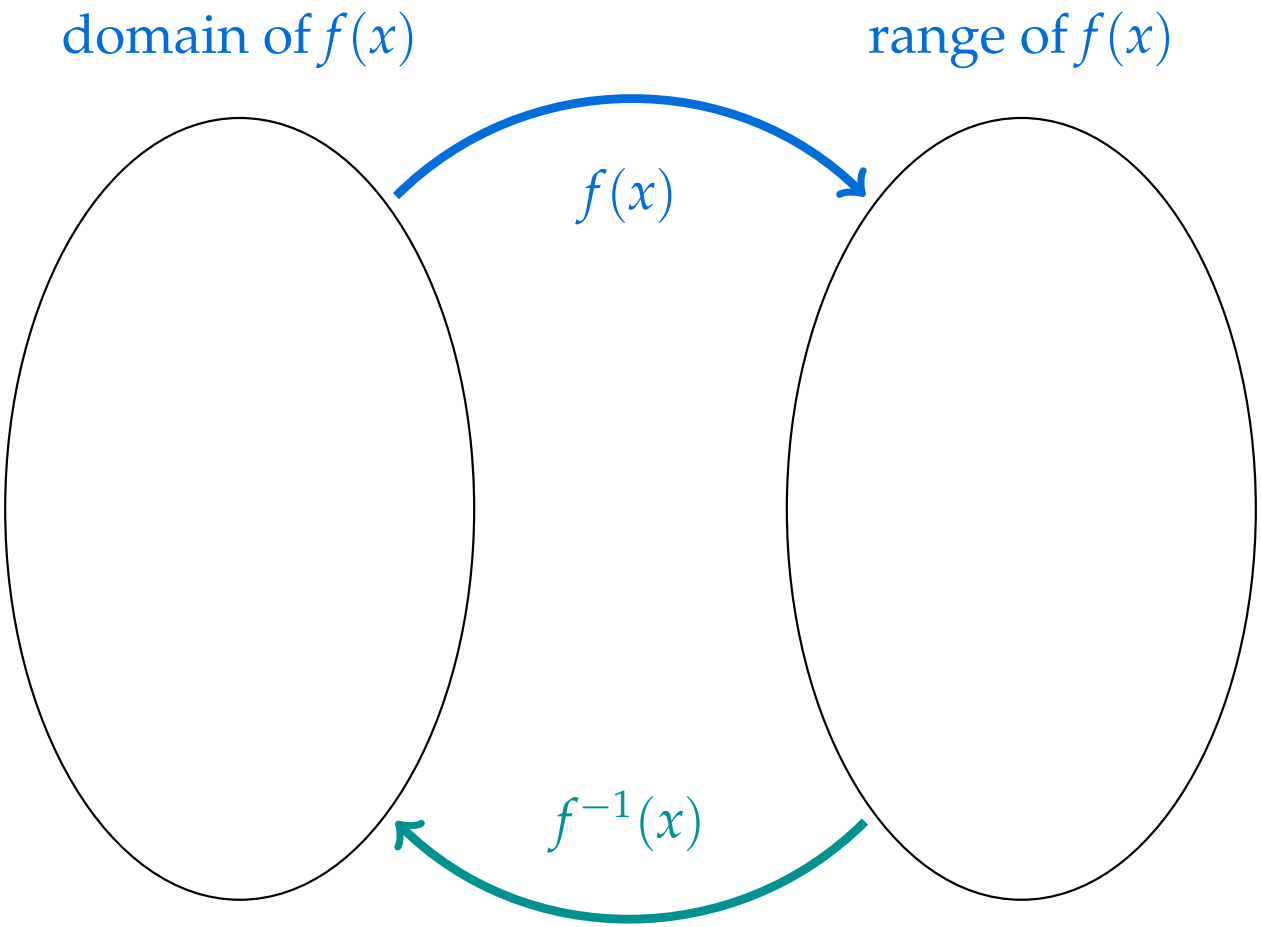

$$f^{-1}(x) = \frac{1+\sqrt{1+4x}}{2}$$

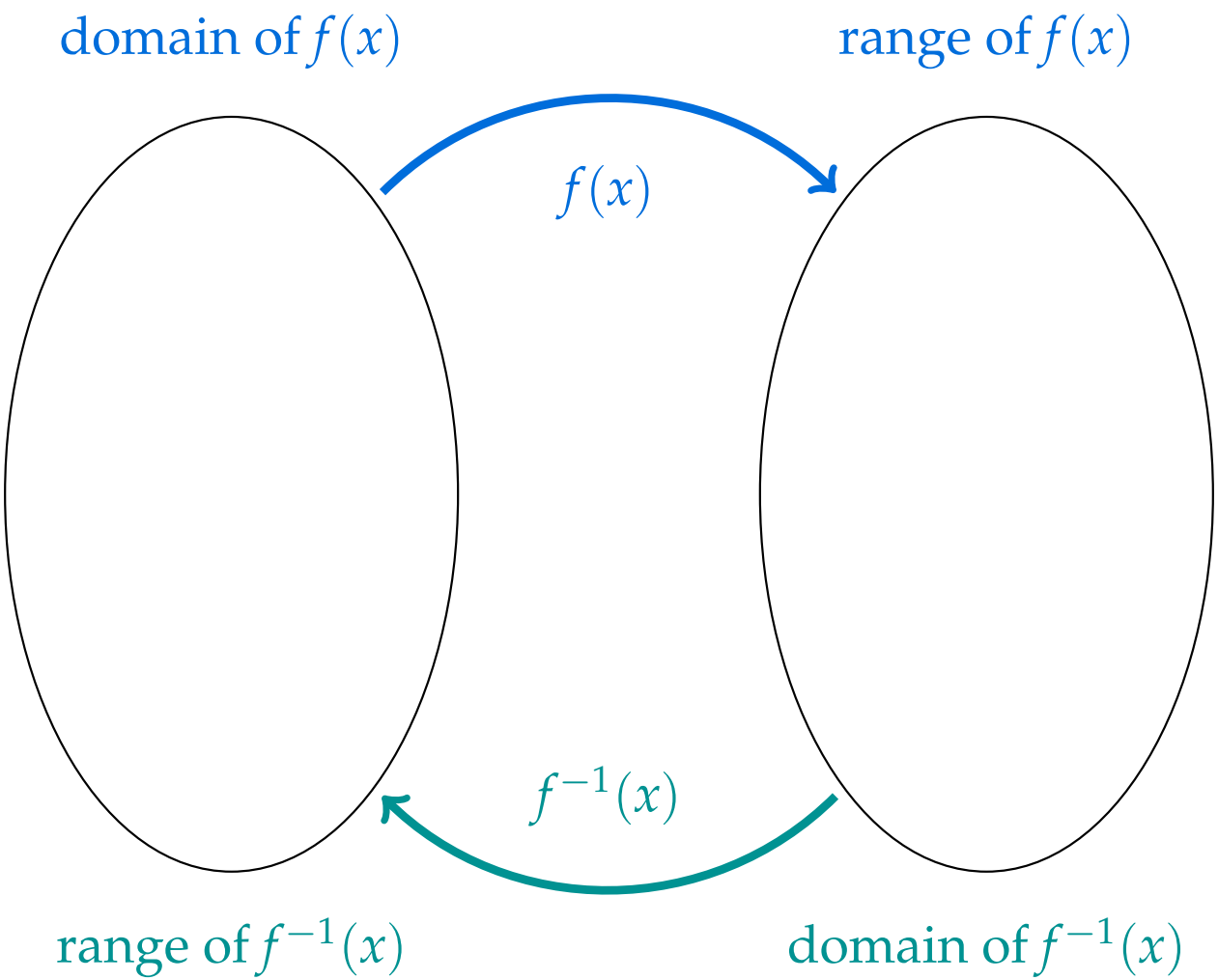
$$a^2 - a = x, \text{ find } a$$

$$a^2 - a - x = 0$$

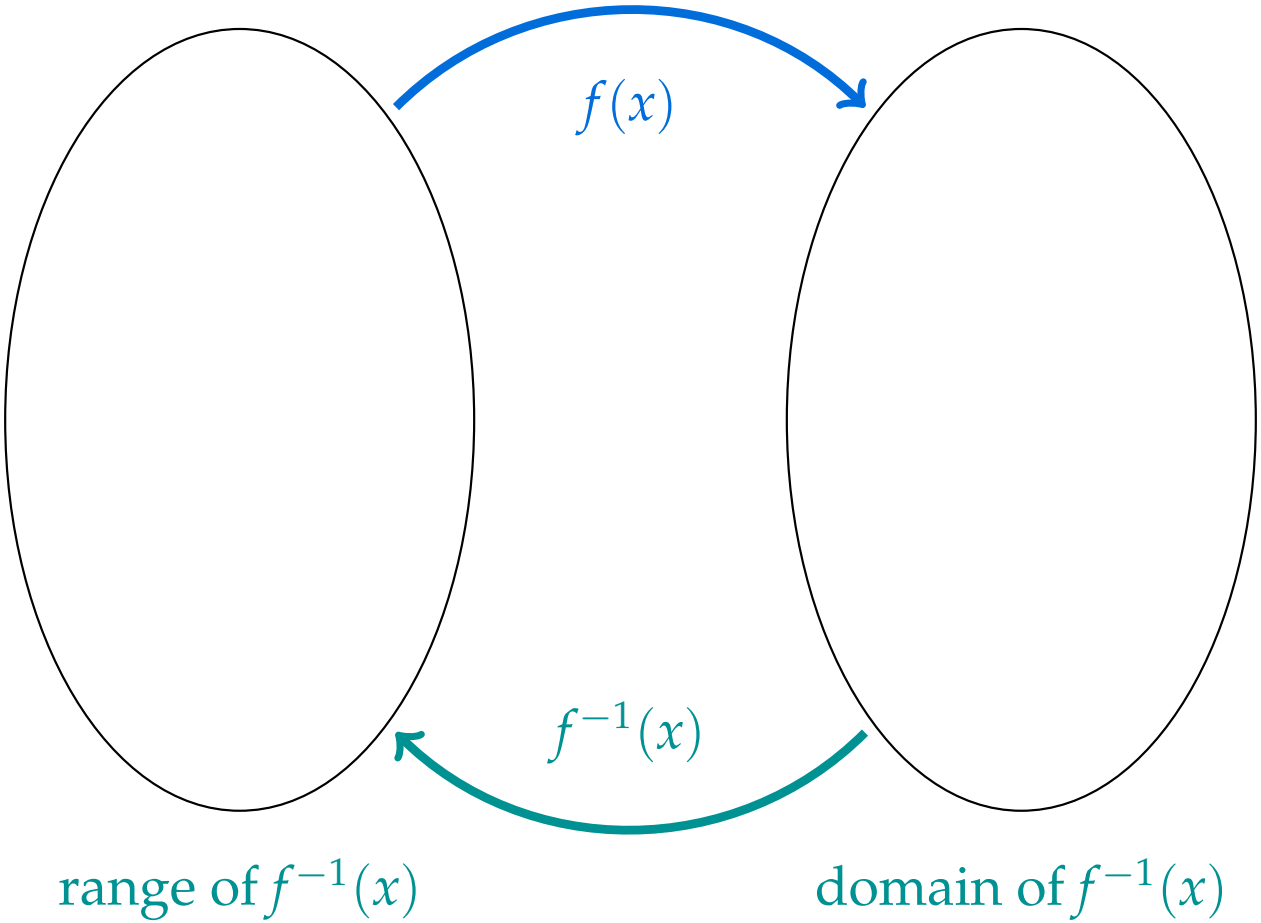
$$a = \frac{1 \pm \sqrt{1+4x}}{2}$$

$$f^{-1}(x) = \frac{1+\sqrt{1+4x}}{2}$$

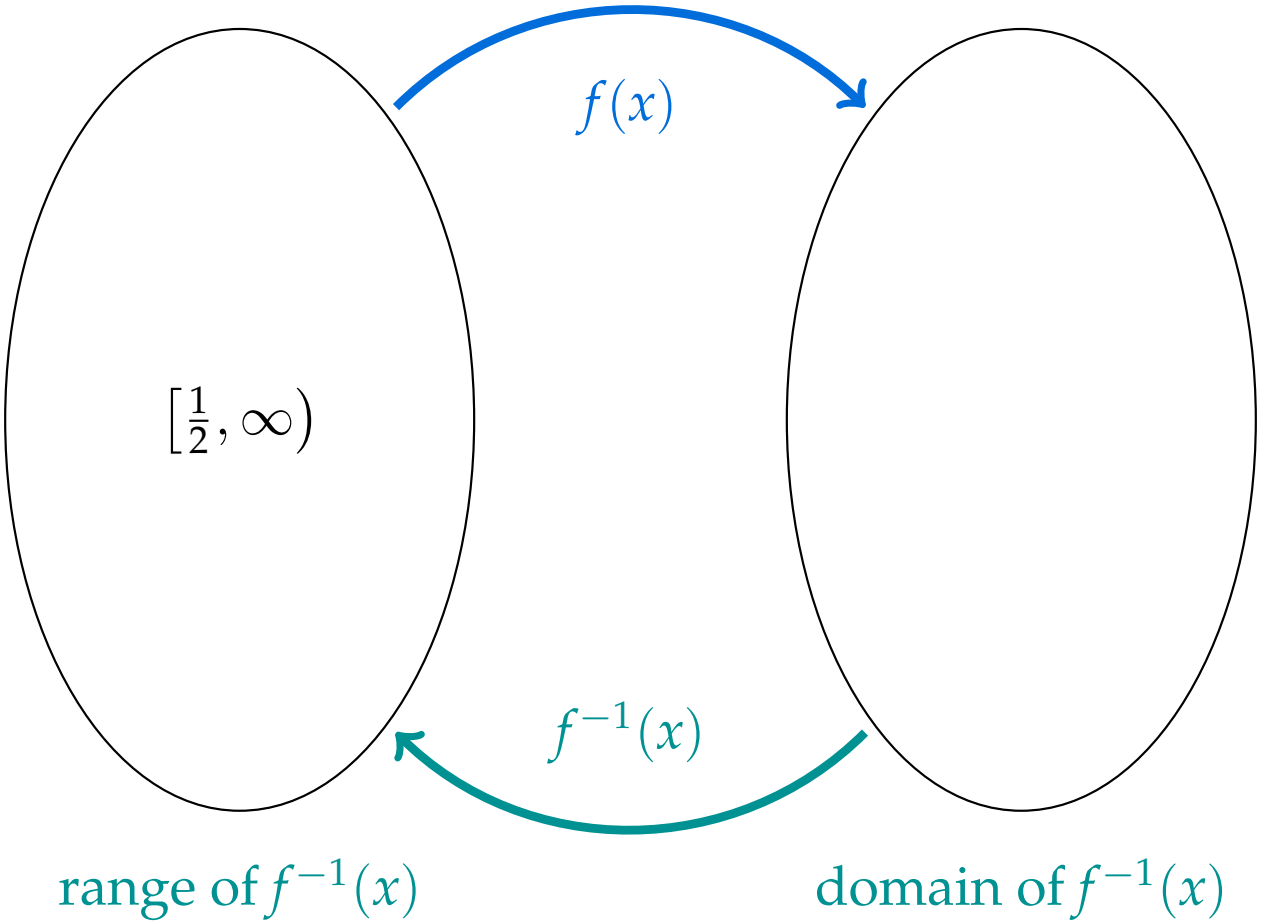

domain of $f(x)$
range of $f(x)$
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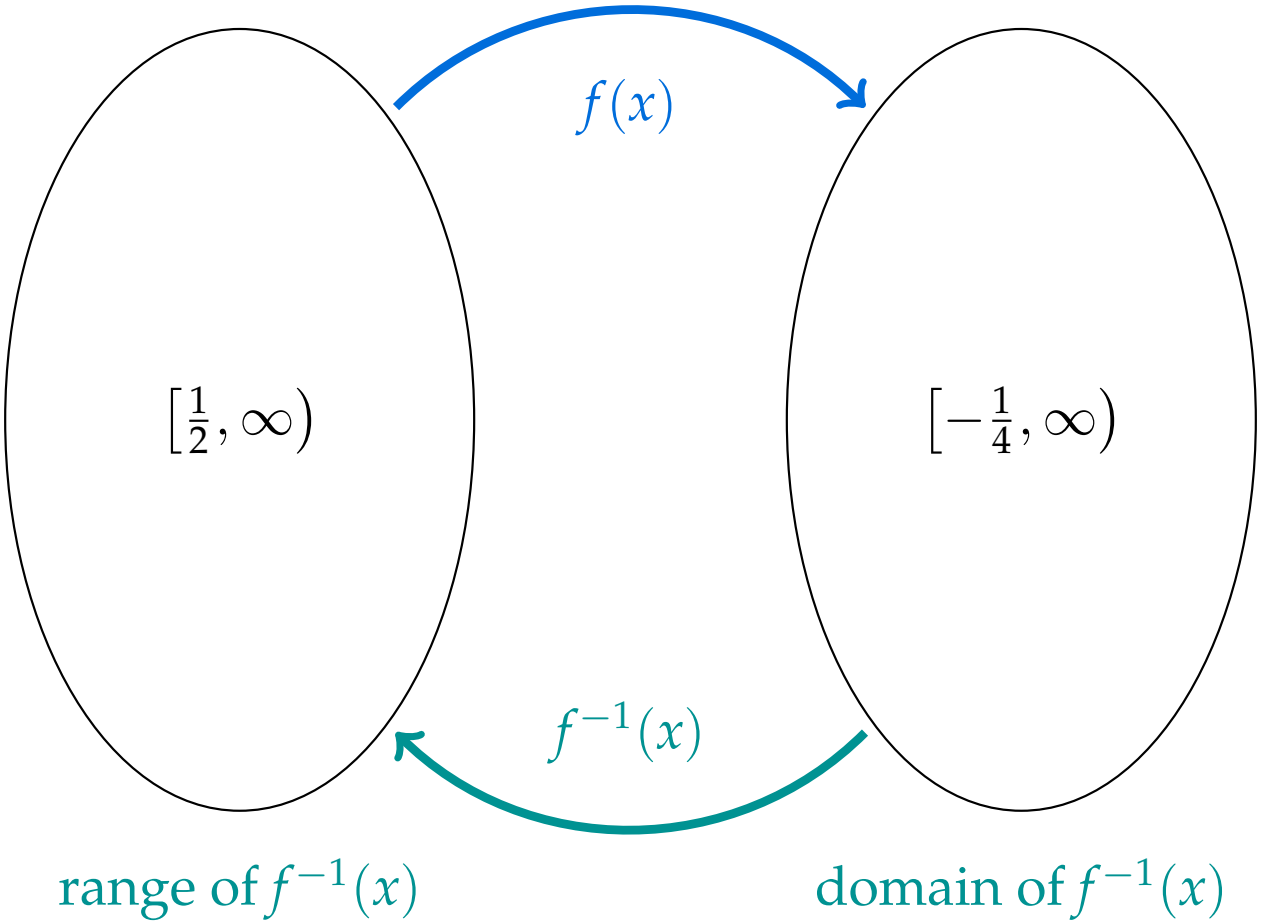
$f(x) = x^2 - x$, domain: $\left[\frac{1}{2}, \infty\right)$



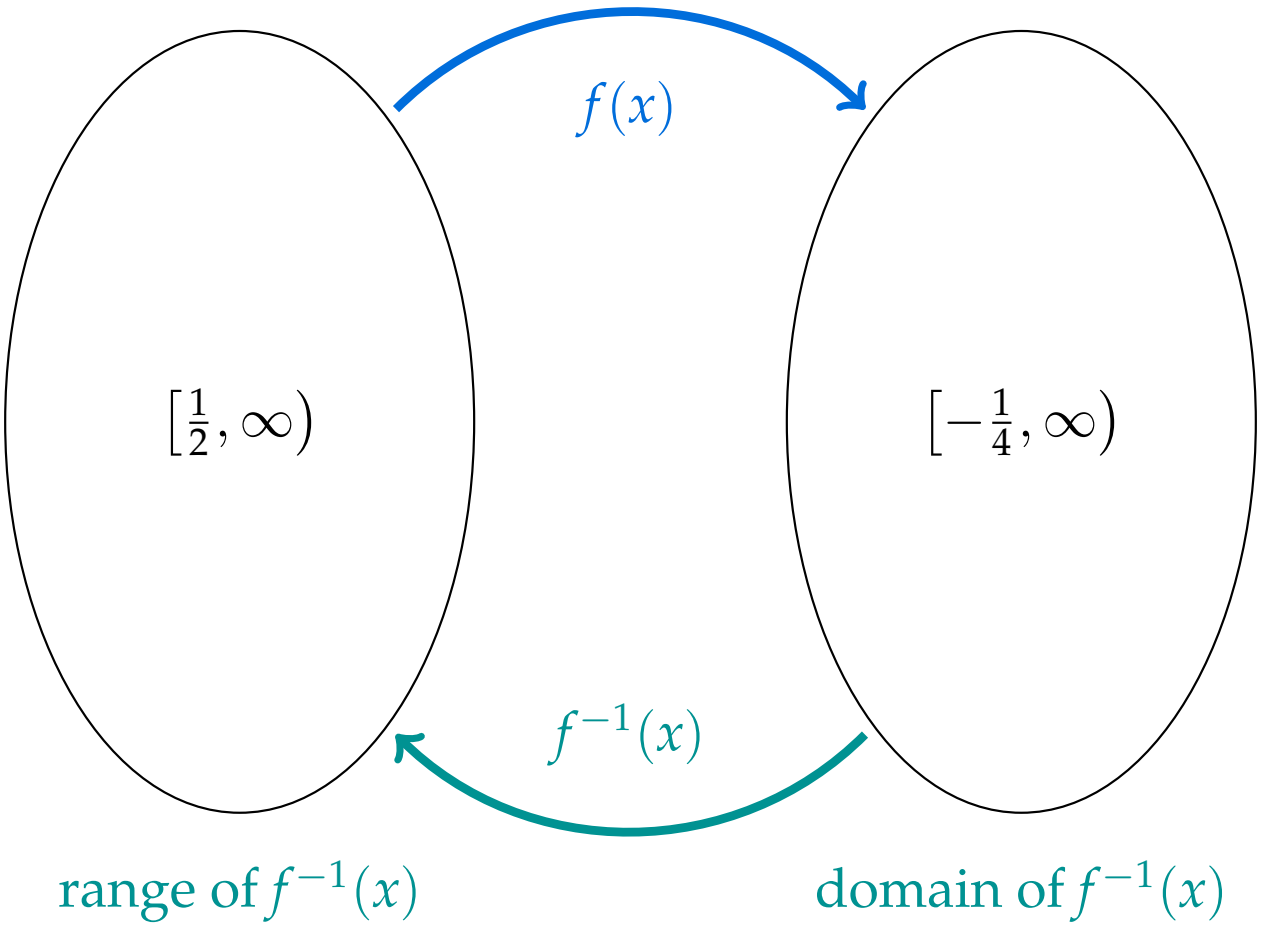
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$f(x) = x^2 - x$, domain: $\left[\frac{1}{2}, \infty\right)$ $\quad f^{-1}(x) = \frac{1+\sqrt{1+4x}}{2}$



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- I'm thinking of an $x$. Your clue: $f(x) = e$. What is my $x$?

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- I'm thinking of an $x$. Your clue: $f(x) = e^3$. What is my $x$? $x = 3$
- I'm thinking of an $x$. Your clue: $f(x) = 0$. What is my $x$? Trick question: no $x$ gives $f(x) = 0$.

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$f^{-1}(x) = \log_e x$

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- I'm thinking of an $x$. Your clue: $f(x) = \frac{1}{e}$. What is my $x$? $x = -1$
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- I'm thinking of an $x$. Your clue: $f(x) = 0$. What is my $x$? Trick question: no $x$ gives $f(x) = 0$.

# INVERTIBILITY GAME: $f(x) = e^x$

$f^{-1}(x) = \log_e x$

- I'm thinking of an $x$. Your clue: $f(x) = e$. What is my $x$? $x = 1$
  $\log_e(e) = 1$
- I'm thinking of an $x$. Your clue: $f(x) = 1$. What is my $x$? $x = 0$
  $\log_e(1) = 0$
- I'm thinking of an $x$. Your clue: $f(x) = \frac{1}{e}$. What is my $x$? $x = -1$
  $\log_e\left(\frac{1}{e}\right) = -1$
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  $\log_e(e^3) = 3$
- I'm thinking of an $x$. Your clue: $f(x) = 0$. What is my $x$? Trick question: no $x$ gives $f(x) = 0$.
  $\log_e(x)$ is undefined at $x = 0$

1. Suppose $0 < x < 1$. Then $\log_e(x)$ is...

2. Suppose $-1 < x < 0$. Then $\log_e(x)$ is...

3. Suppose $e < x$. Then $\log_e(x)$ is...

A. positive
B. negative
C. greater than one
D. less than one
E. undefined

# Exponents and Logarithms

$$f(x) = e^x \qquad f^{-1}(x) = \log_e(x) = \ln(x) = \log(x)$$

# Exponents and Logarithms

$f(x) = e^x$ $\qquad$ $f^{-1}(x) = \log_e(x) = \ln(x) = \log(x)$

| $x$ | $e^x$ |
|---|---|
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| 1 | $e$ |
| $-1$ | $\frac{1}{e}$ |
| $n$ | $e^n$ |

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| $0$ | $1$ | $e^0 = 1 \leftrightarrow \log_e(1) = 0$ | $1$ | $0$ |
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# Exponents and Logarithms

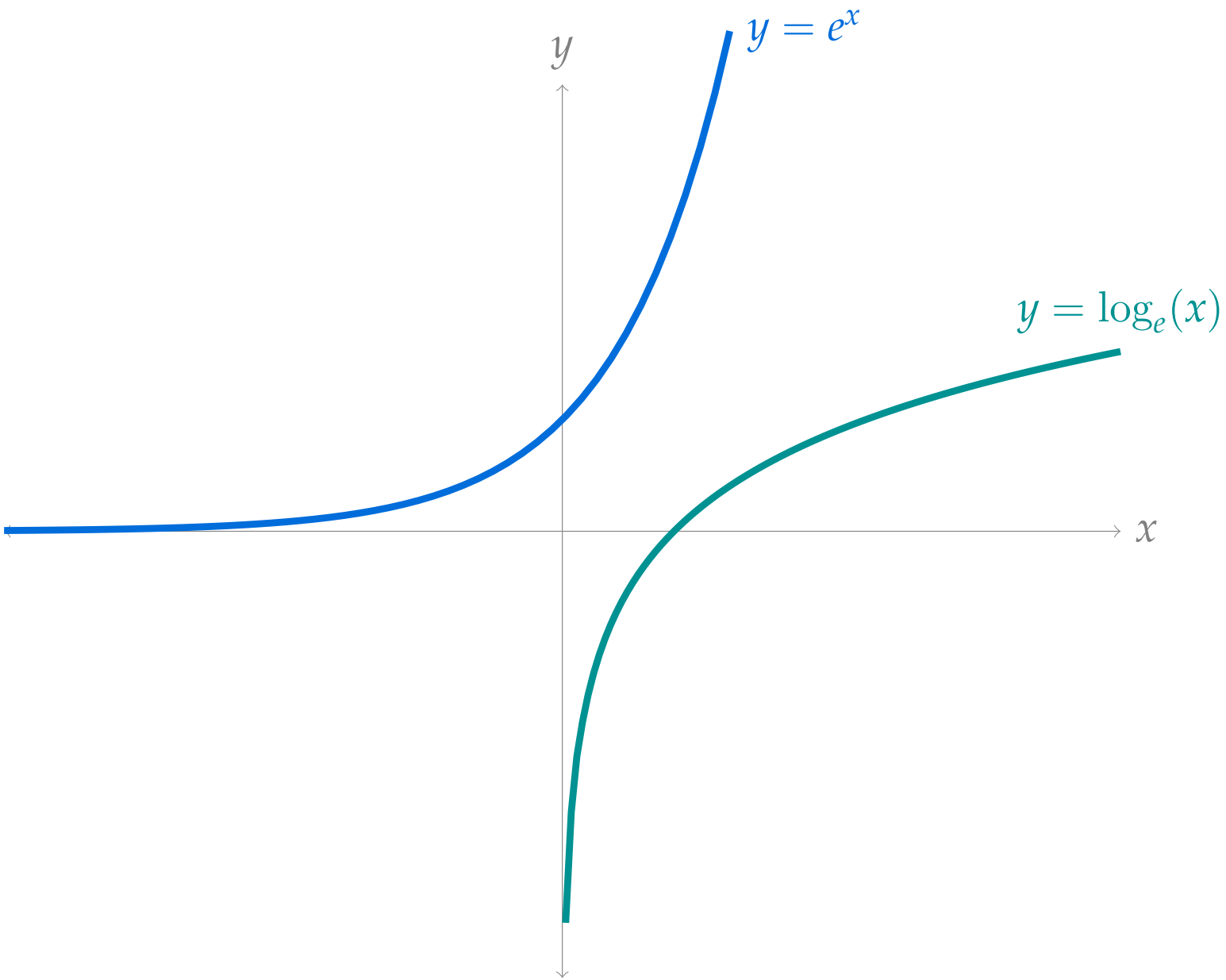
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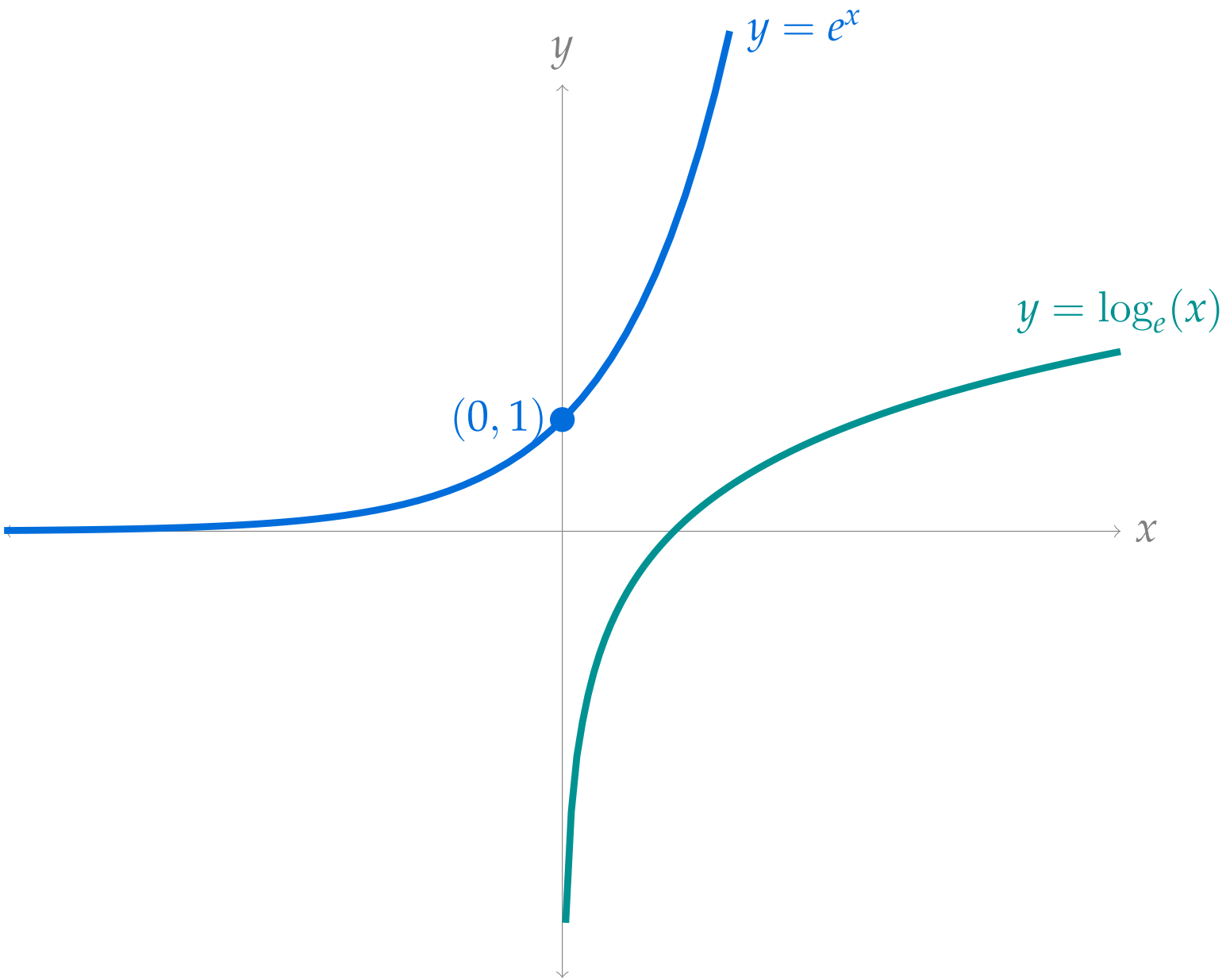
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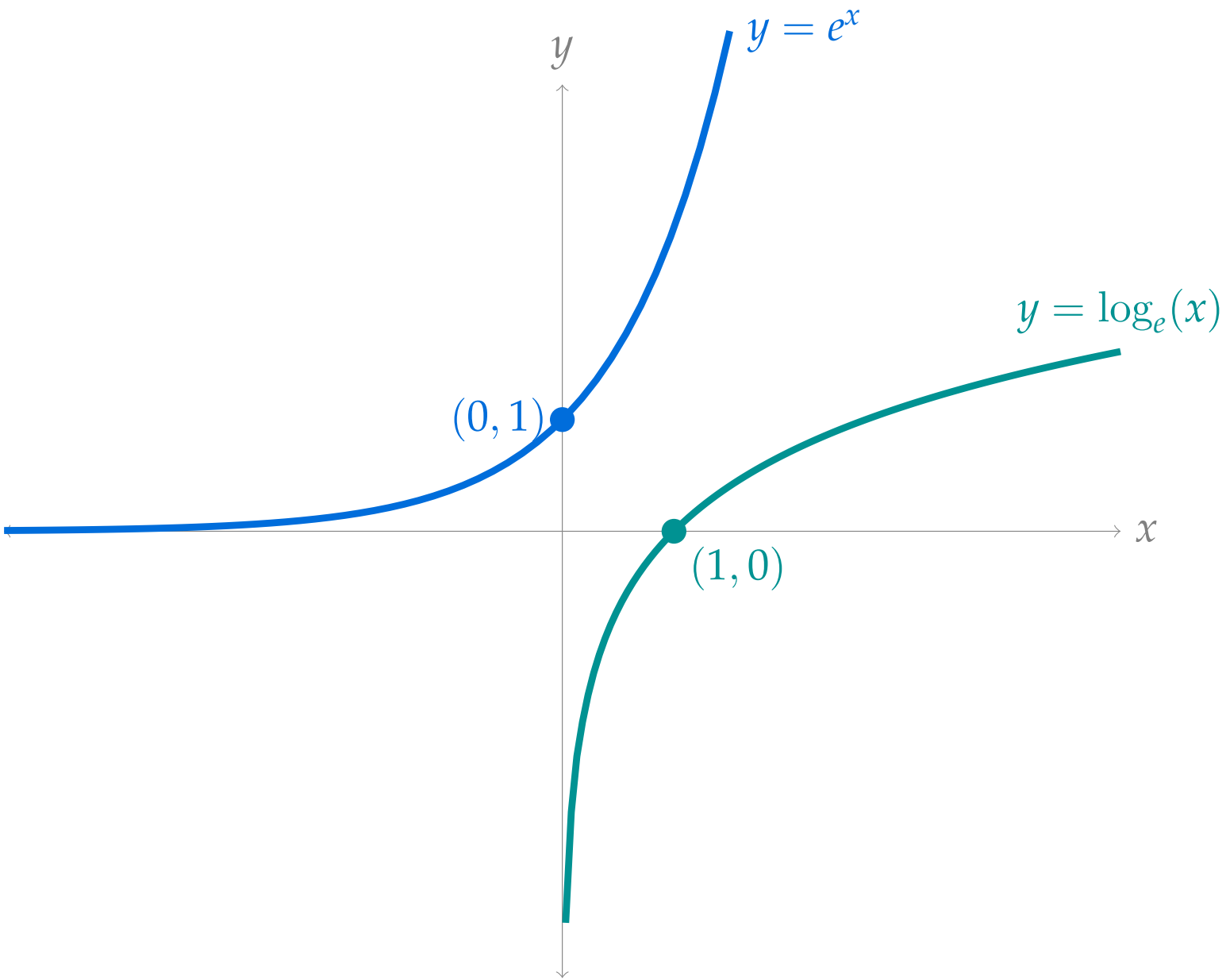
# Exponents and Logarithms

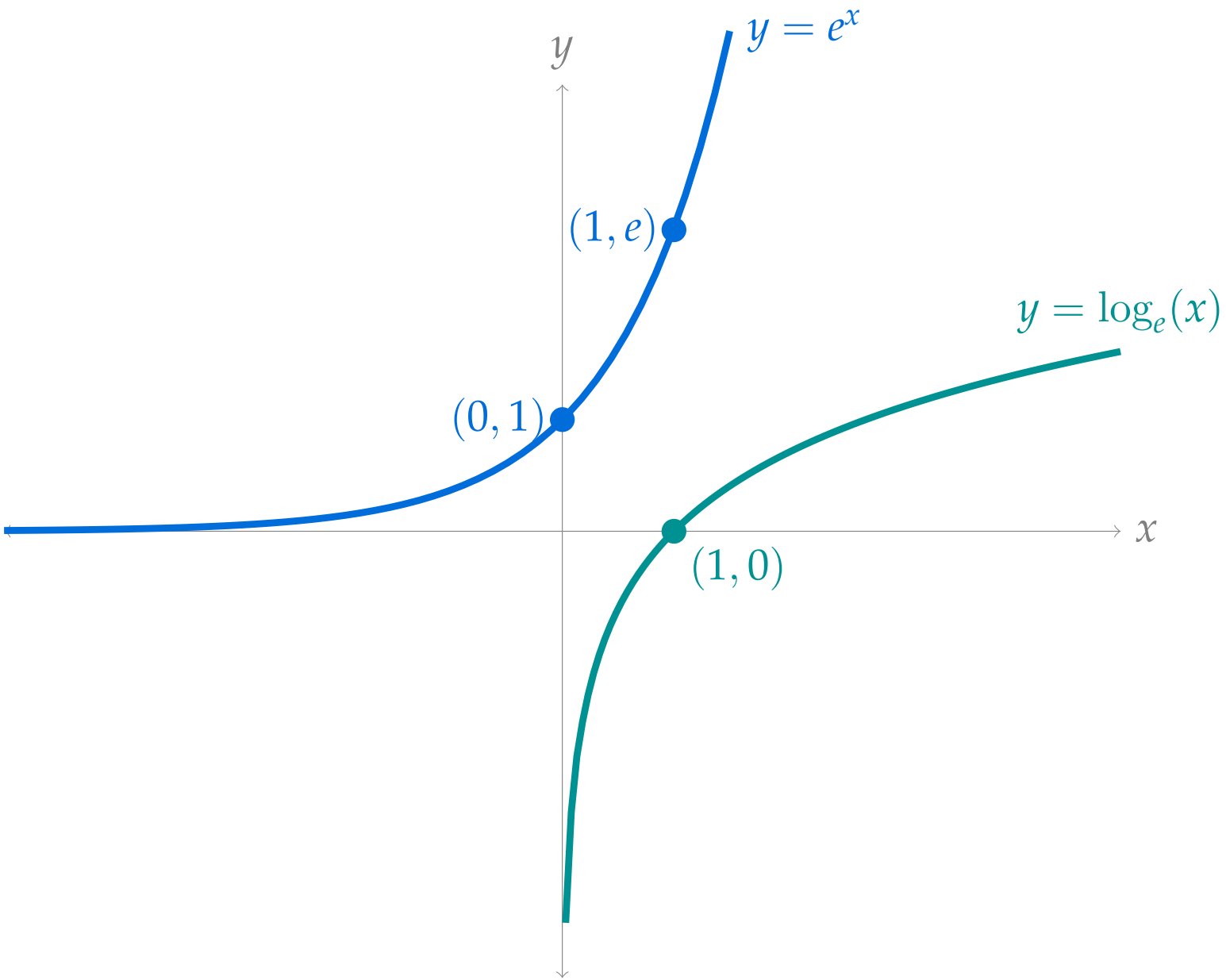
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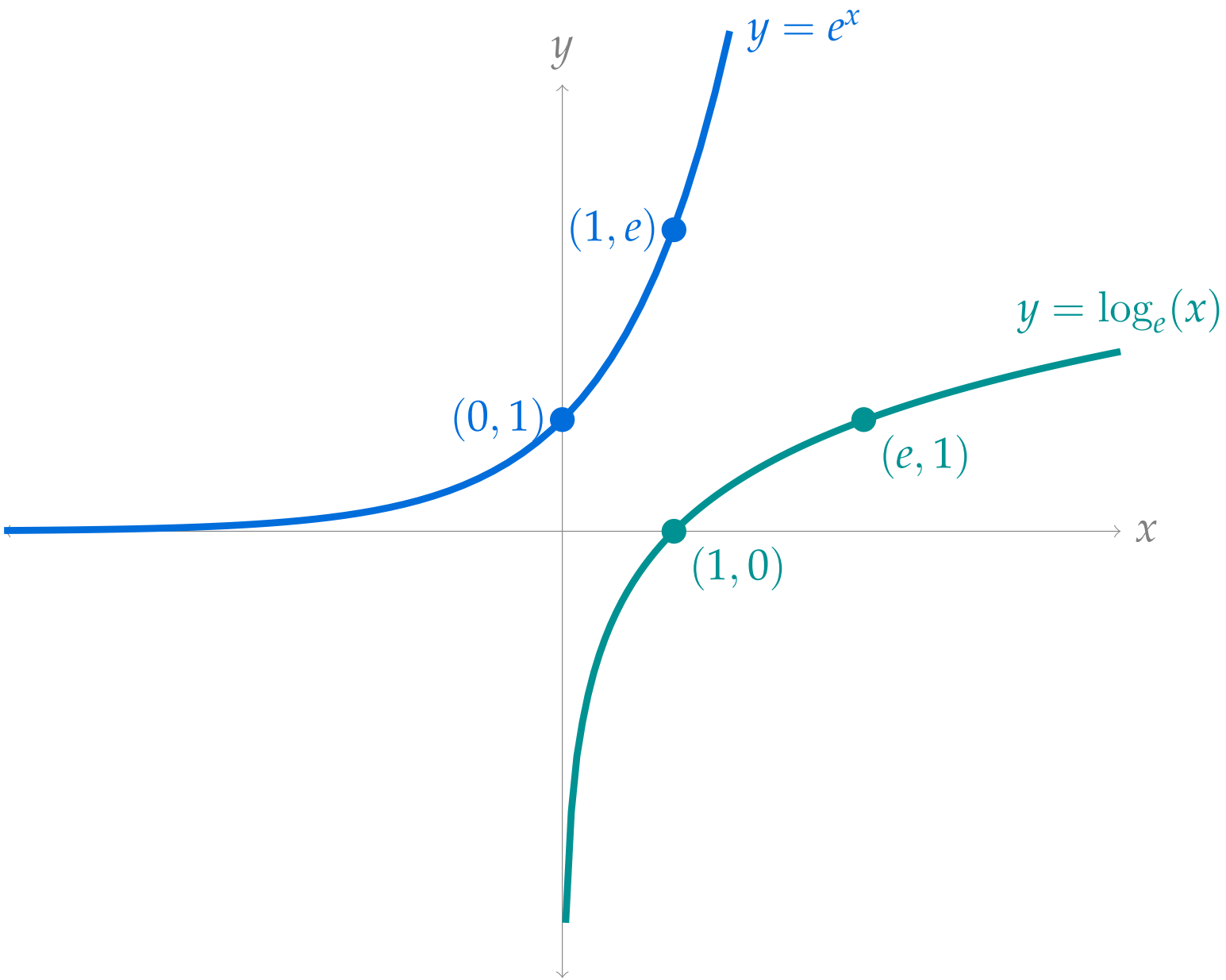
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| $n$ | $e^n$ | $e^n = e^n \leftrightarrow \log_e(e^n) = n$ | $e^n$ | $n$ |

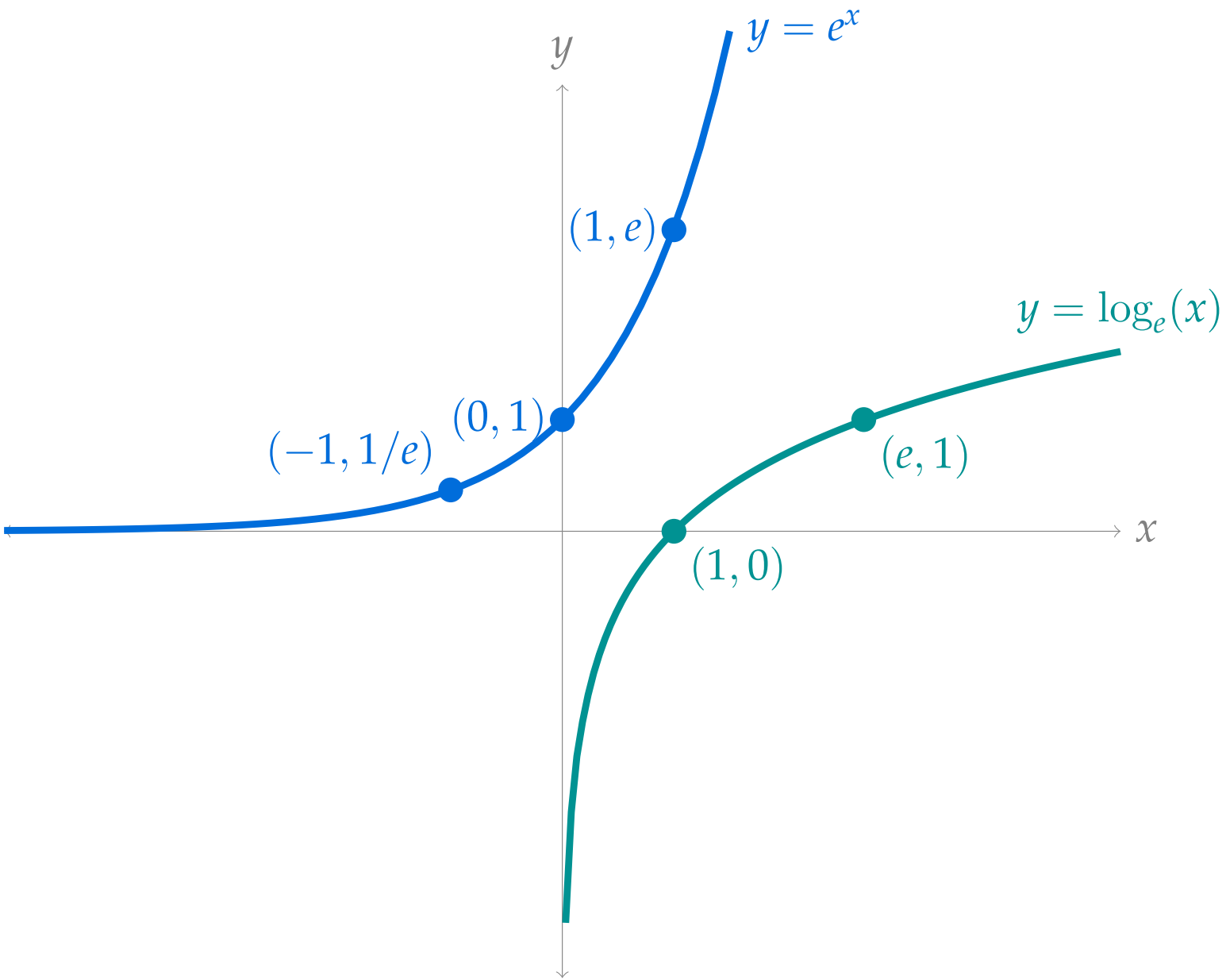

$y = e^x$
$y = \log_e(x)$
$y$
$x$

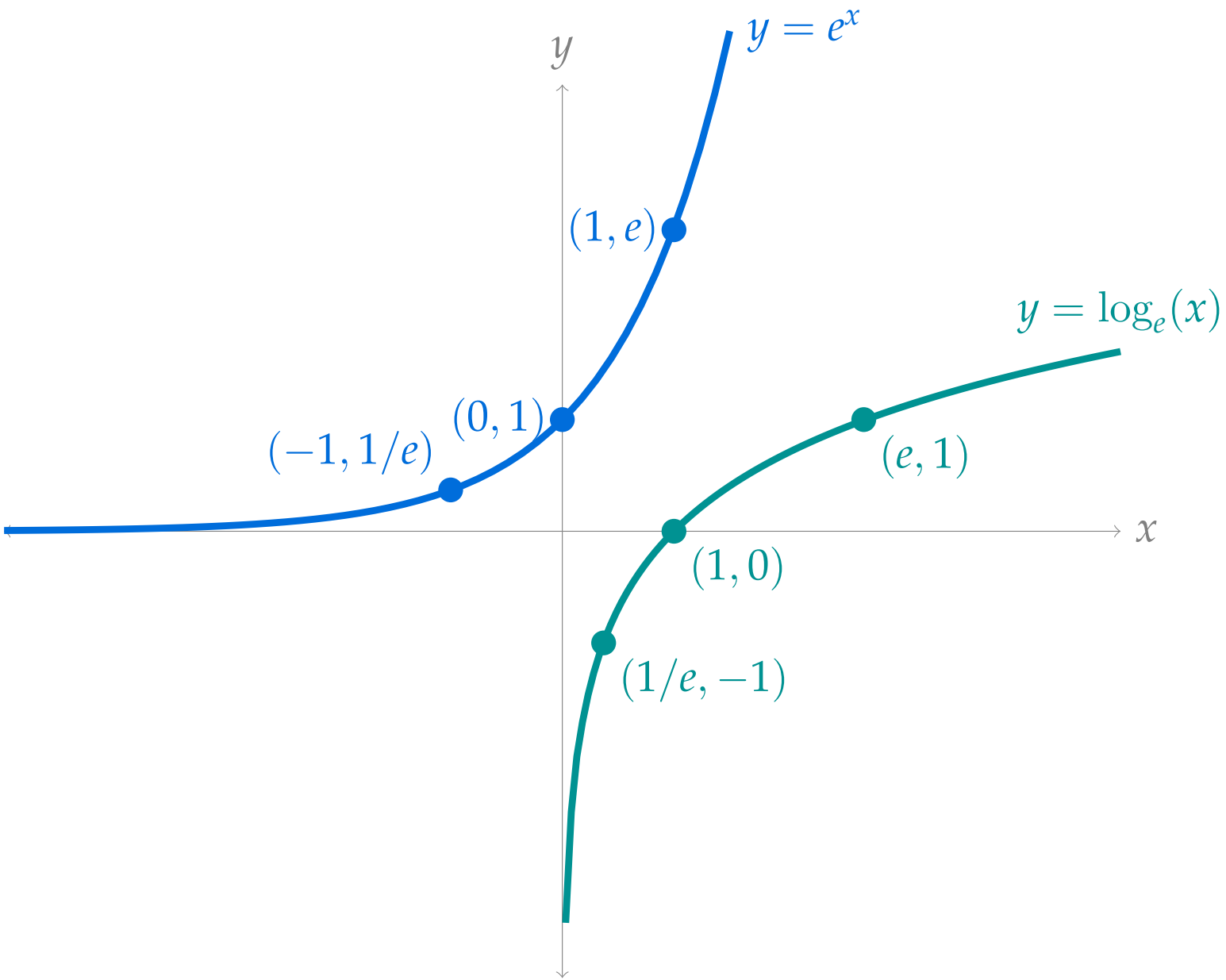

$y = e^x$
$y$
$y = \log_e(x)$
$(0, 1)$
$x$

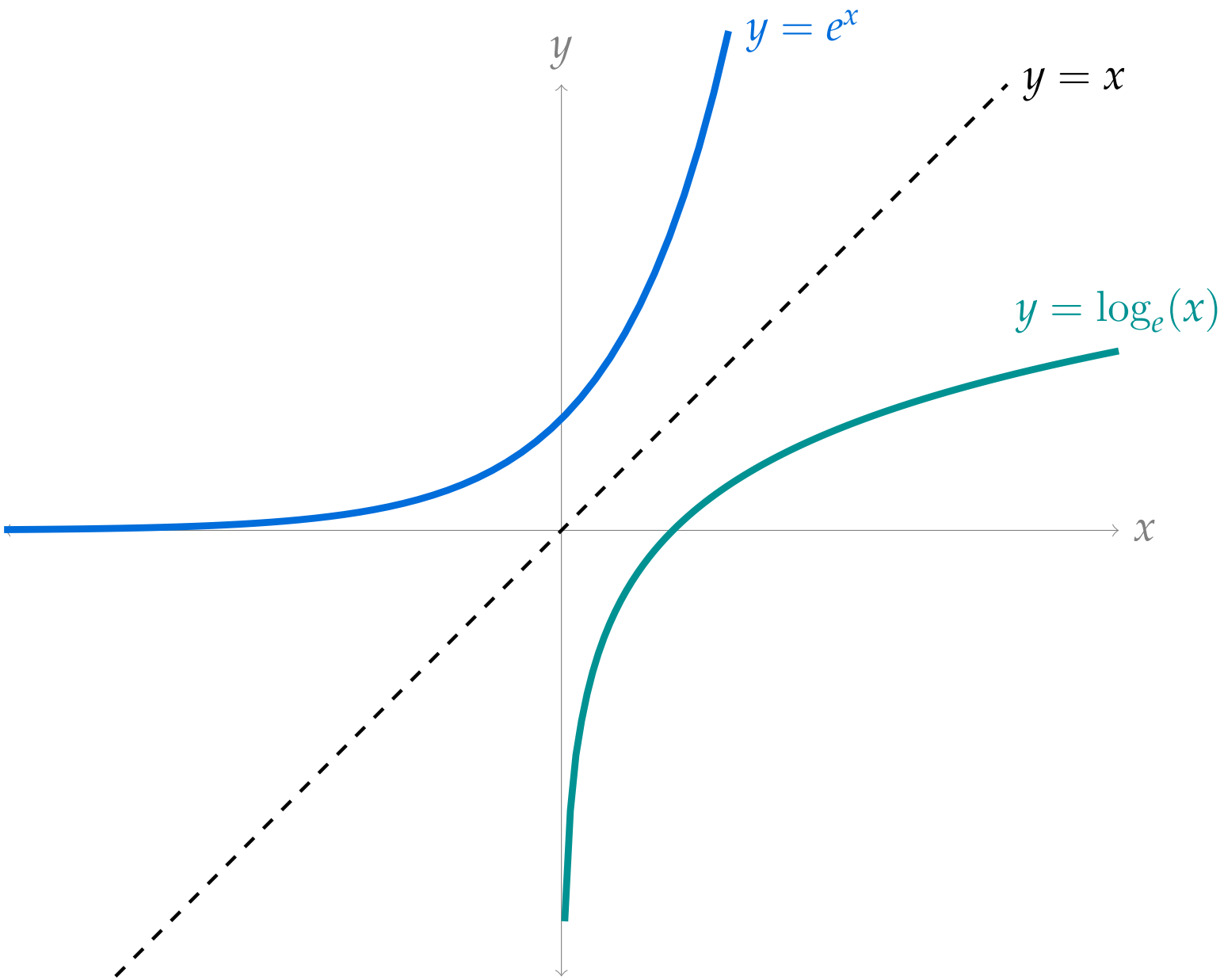

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$(0, 1)$
$x$
$(1, 0)$


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$x$
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$(1/e, -1)$


$y = e^x$
$y = x$
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$y$
$x$

# Logs of Other Bases: $\log_n(x)$ is the inverse of $n^x$

$\log_{10} 10^8 =$

A. 0

B. 8

C. 10

D. other

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$\log_2 16 =$

A. 1

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C. 3

D. other ✓ $2^4 = 16$ so $\log_2 16 = 4$

## Logarithm Rules

Let $A$ and $B$ be positive, and let $n$ be any real number.

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$\log(A \cdot B) = \log(A) + \log(B)$

Proof: $\log(A \cdot B) = \log(e^{\log A} e^{\log B}) = \log(e^{\log A + \log B}) = \log(A) + \log(B)$

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Write as a single logarithm:

$$f(x) = \log\left(\frac{10}{x^2}\right) + 2\log x + \log(10 + x)$$

## Logarithm Rules

Let $A$ and $B$ be positive, and let $n$ be any real number.

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$$\begin{aligned} f(x) &= \log\left(\tfrac{10}{x^2}\right) + 2\log x + \log(10+x) \\ &= \log 10 - \log(x^2) + 2\log x + \log(10+x) \\ &= \log 10 - 2\log x + 2\log x + \log(10+x) \\ &= \log 10 + \log(10+x) = \log(10(10+x)) \\ &= \log(100+10x) \end{aligned}$$

# Base Change

Fact: $b^{\log_b(a)} = a$

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$$\Rightarrow \log\left(b^{\log_b(a)}\right) = \log(a)$$

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$$\Rightarrow \log_b(a) = \frac{\log(a)}{\log(b)}$$

# Base Change

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Suppose your calculator can only compute logarithms base 10. What would you enter to calculate $\log(17)$?

Suppose your calculator can only compute natural logarithms. What would you enter to calculate $\log_2(57)$?

Suppose your calculator can only compute logarithms base 2. What would you enter to calculate $\log(2)$?

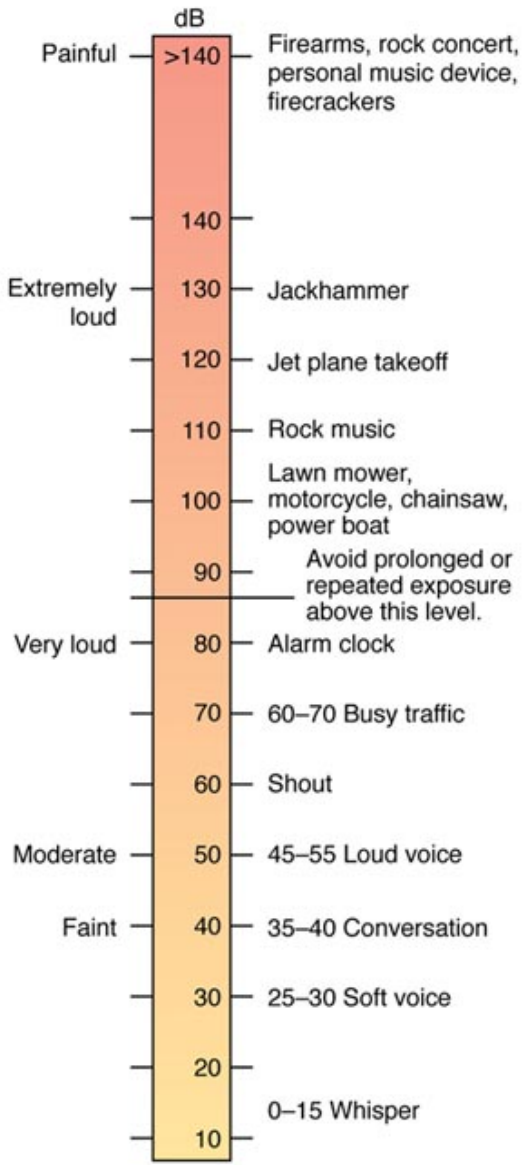
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Suppose your calculator can only compute logarithms base 2. What would you enter to calculate $\log(2)$? $\frac{\log_2 2}{\log_2 e} = \frac{1}{\log_2 e}$




Decibels: For a particular measure of the power $P$ of a sound wave, the decibels of that sound is:

$$10 \log_{10}(P)$$

So, every ten decibels corresponds to a sound being ten **times** louder.

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A. 100 dB

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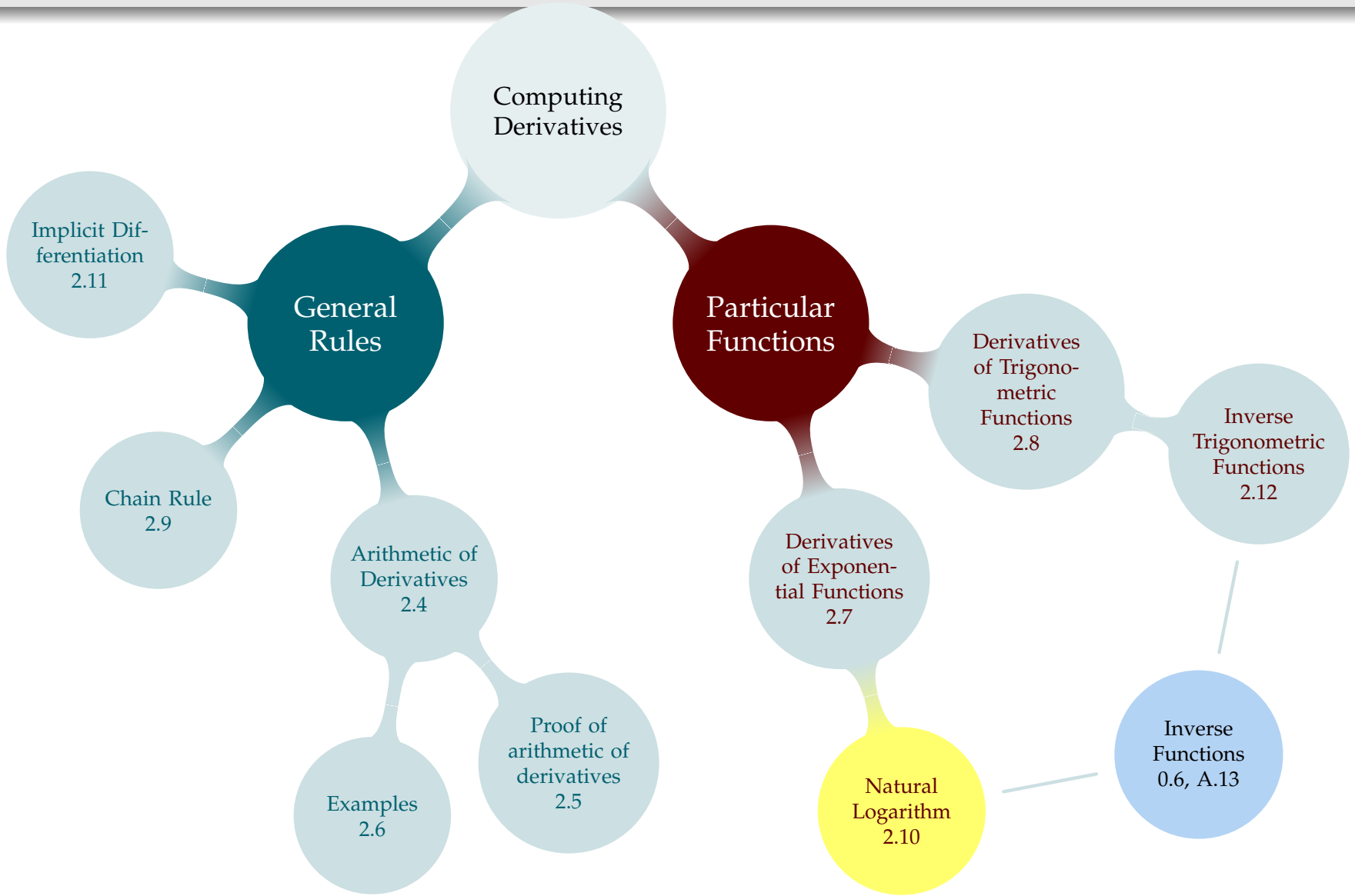
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# Table of Contents



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$$1 = e^{\log_e x} \cdot \frac{\mathrm{d}}{\mathrm{d}x}\{\log_e x\} = x \cdot \frac{\mathrm{d}}{\mathrm{d}x}\{\log_e x\}$$

$$\frac{1}{x} = \frac{\mathrm{d}}{\mathrm{d}x}\{\log_e x\}$$

# Differentiating the Natural Logarithm

Calculate $\frac{\mathrm{d}}{\mathrm{d}x}\{\log_e x\}$.

## Derivative of Natural Logarithm – Theorem 2.10.1

$$\frac{\mathrm{d}}{\mathrm{d}x}\{\log_e x\} = \frac{1}{x} \qquad (x > 0)$$

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We use the chain rule:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left\{\log_e\left|\boxed{x^2+1}\right|\right\} = \frac{1}{x^2+1}\cdot(2x)$$
$$= \frac{2x}{x^2+1}$$

## Derivatives of Logarithms – Corollary 2.10.6

For $a > 0$:

$$\frac{\mathrm{d}}{\mathrm{d}x}[\log_a |x|] = \frac{1}{x \log a}$$

In particular:

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Differentiate: $f(x) = \log_e |\cot x|$

We use the chain rule:

$$\frac{\mathrm{d}}{\mathrm{d}x}[\log_e \boxed{|\cot x|}] = \frac{1}{\cot x} \cdot (-\csc^2 x) = \frac{-\csc^2 x}{\cot x}$$

# Logarithmic Differentiation - A Fancy Trick

- $\log(f \cdot g) = \log f + \log g$
- $\log\left(\frac{f}{g}\right) = \log f - \log g$
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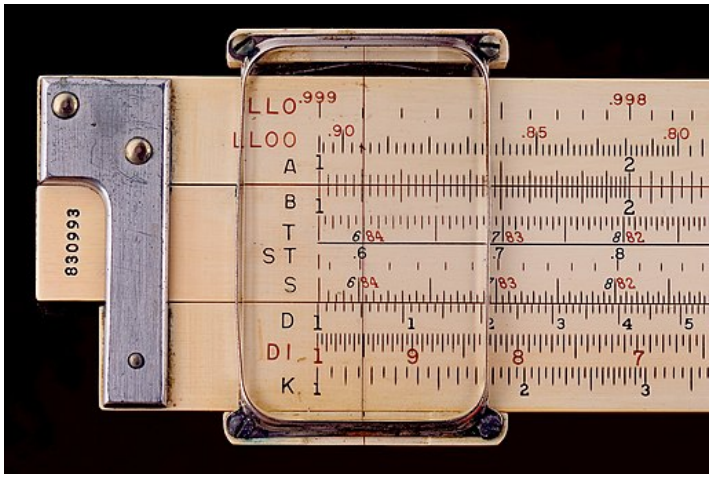
  We can exploit these properties to differentiate!

## Logarithmic Differentiation

In general, if $f(x) \neq 0$, $\frac{\mathrm{d}}{\mathrm{d}x}\left[\log|f(x)|\right] = \frac{f'(x)}{f(x)}$.

$$f(x) = \left(\frac{(2x+5)^4(x^2+1)}{x+3}\right)^5$$

Find $f'(x)$.



# Logarithmic Differentiation - A Fancy Trick

$$f(x) = \left( \frac{(2x+5)^4(x^2+1)}{x+3} \right)^5$$

# Logarithmic Differentiation - A Fancy Trick

$$f(x) = \left(\frac{(2x+5)^4(x^2+1)}{x+3}\right)^5$$

$$\log(f(x)) = \log\left[\left(\frac{(2x+5)^4(x^2+1)}{x+3}\right)^5\right]$$

$$= 5\log\left[\frac{(2x+5)^4(x^2+1)}{x+3}\right]$$

$$= 5\Big[4\log(2x+5) + \log(x^2+1) - \log(x+3)\Big]$$

$$\frac{f'(x)}{f(x)} = 5\left[4\frac{2}{2x+5} + \frac{2x}{x^2+1} - \frac{1}{x+3}\right]$$

$$f'(x) = \left(\frac{(2x+5)^4(x^2+1)}{x+3}\right)^5 \cdot 5\left[4\frac{2}{2x+5} + \frac{2x}{x^2+1} - \frac{1}{x+3}\right]$$

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# Logarithmic Differentiation - A Fancy Trick

Differentiate:

$$f(x) = x^x$$

$$\begin{aligned}
\log(f(x)) &= \log\left[x^x\right] \\
&= x \log x \\
\frac{f'(x)}{f(x)} &= x \cdot \frac{1}{x} + \log x \cdot 1 \\
&= 1 + \log x \\
f'(x) &= x^x \left[1 + \log x\right]
\end{aligned}$$

# Logarithmic Differentiation - A Fancy Trick

Differentiate:

$$f(x) = \left( \frac{(x^{15} - 9x^2)^{10}(x + x^2 + 1)}{(x^7 + 7)(x + 1)(x + 2)(x + 3)} \right)^5$$

# Logarithmic Differentiation - A Fancy Trick

Differentiate:

$$f(x) = \left( \frac{(x^{15} - 9x^2)^{10}(x + x^2 + 1)}{(x^7 + 7)(x + 1)(x + 2)(x + 3)} \right)^5$$

$$\begin{aligned}
\log(f(x)) &= \log\left[ \left( \frac{(x^{15} - 9x^2)^{10}(x + x^2 + 1)}{(x^7 + 7)(x + 1)(x + 2)(x + 3)} \right)^5 \right] \\
&= 5 \log\left[ \frac{(x^{15} - 9x^2)^{10}(x + x^2 + 1)}{(x^7 + 7)(x + 1)(x + 2)(x + 3)} \right] \\
&= 5\Big[10 \log\left(x^{15} - 9x^2\right) + \log(x + x^2 + 1) - \log(x^7 + 7) - \log(x + 1) - \log(x + 2) - \log(x + 3)\Big] \\
\frac{f'(x)}{f(x)} &= 5\left[10 \frac{15x^{14} - 18x}{x^{15} - 9x^2} + \frac{1 + 2x}{x + x^2 + 1} - \frac{7x^6}{x^7 + 7} - \frac{1}{x + 1} - \frac{1}{x + 2} - \frac{1}{x + 3}\right] \\
f'(x) &= \left( \frac{(x^{15} - 9x^2)^{10}(x + x^2 + 1)}{(x^7 + 7)(x + 1)(x + 2)(x + 3)} \right)^5 \cdot 5\left[10 \frac{15x^{14} - 18x}{x^{15} - 9x^2} + \frac{1 + 2x}{x + x^2 + 1} - \frac{7x^6}{x^7 + 7} - \frac{1}{x + 1} - \frac{1}{x + 2} - \frac{1}{x + 3}\right]
\end{aligned}$$

$$f(x) = \frac{(x^8 - e^x)(\sqrt{x} + 5)}{\csc^5 x}$$

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$$\frac{(x^8 - e^x)(\sqrt{x} + 5)}{\csc^5 x} = f(x)$$

$$\log \left| \frac{(x^8 - e^x)(\sqrt{x} + 5)}{\csc^5 x} \right| = \log |f(x)|$$

$$\left[ \log |x^8 - e^x| + \log |x^{1/2} + 5| - 5 \log |\csc x| \right] = \log |f(x)|$$

$$\frac{\mathrm{d}}{\mathrm{d}x} \left[ \log |x^8 - e^x| + \log |x^{1/2} + 5| - 5 \log |\csc x| \right] = \frac{\mathrm{d}}{\mathrm{d}x} \log |f(x)|$$

$$\frac{8x^7 - e^x}{x^8 - e^x} + \frac{\frac{1}{2}x^{-1/2}}{x^{1/2} + 5} - 5 \frac{-\csc x \cot x}{\csc x} = \frac{f'(x)}{f(x)}$$

$$\frac{(x^8 - e^x)(\sqrt{x} + 5)}{\csc^5 x} \left( \frac{8x^7 - e^x}{x^8 - e^x} + \frac{\frac{1}{2}x^{-1/2}}{x^{1/2} + 5} + 5 \frac{\csc x \cot x}{\csc x} \right) = f'(x)$$

$$f(x) = (x^2 + 17)(32x^5 - 8)(x^{98} - x^{57} + 32x^2)^4(32x^{10} - 10x^{32})$$

Find $f'(x)$.

$$f(x) = (x^2+17)(32x^5-8)(x^{98}-x^{57}+32x^2)^4(32x^{10}-10x^{32})$$

Find $f'(x)$.

$$(x^2+17)(32x^5-8)(x^{98}-x^{57}+32x^2)^4(32x^{10}-10x^{32}) = f(x)$$

$$\log\left|(x^2+17)(32x^5-8)(x^{98}-x^{57}+32x^2)^4(32x^{10}-10x^{32})\right| = \log|f(x)|$$

$$\log|x^2+17| + \log|32x^5-8| + 4\log|x^{98}-x^{57}+32x^2| + \log|32x^{10}-10x^{32}| = \log|f(x)|$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[\log|x^2+17| + \log|32x^5-8| + 4\log|x^{98}-x^{57}+32x^2| + \log|32x^{10}-10x^{32}|\right] = \frac{\mathrm{d}}{\mathrm{d}x}[\log|f(x)|]$$

$$\frac{2x}{x^2+17} + \frac{160x^4}{32x^5-8} + 4\frac{98x^{97}-57x^{56}+64x}{x^{98}-x^{57}+32x^2} + \frac{320x^9-320x^{31}}{32x^{10}-10x^{32}} = \frac{f'(x)}{f(x)}$$

$$\left((x^2+17)(32x^5-8)(x^{98}-x^{57}+32x^2)^4(32x^{10}-10x^{32})\right)\cdot$$

$$\left(\frac{2x}{x^2+17} + \frac{160x^4}{32x^5-8} + 4\frac{98x^{97}-57x^{56}+64x}{x^{98}-x^{57}+32x^2} + \frac{320x^9-320x^{31}}{32x^{10}-10x^{32}}\right) = f'(x)$$

A

# Included Work



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